TECHNICAL REPORT • REPRODUCIBLE RESEARCH ARTEFACT • v1.7

SPHERExLens

A Reproducible Differential-Imaging Pipeline for the SPHEREx Quick-Release Data, with an Honest Triage Layer

Question. Given only public Level-2 spectral images from NASA's SPHEREx all-sky survey, can a single workstation detect, classify and triage *what changed* between two epochs – and can it say honestly when it has found nothing?

Answer. Yes: a six-stage pipeline (ingest → co-add → proper subtraction → catalogue triage → spectral diagnosis → injection–recovery), served by a browser dashboard over 18 fields, with every number traceable to an archive record.

Artefact	`step1_ingest.py` … `step6_injection.py`, `app.py`, `static/`, `site/`
Data source	NASA/IPAC IRSA, SPHEREx Level-2 (`spherex_qr2`), SIA2 + cutout service
Catalogues	Gaia DR3 (TAP), AllWISE, AAVSO VSX (VizieR), 5000-row caps
Fields	18 (5 brief targets + 13 famous objects); 128 cutouts; 0.88 GB
Detector/sky	180×180 px TAN, $3''\,\mathrm{px}^{-1}$, $9'$ field
Headline result	88 detections over 18 fields: 71 KNOWN, 17 CANDIDATE; 4 fields yield nothing
Document	Part I: report • Part II: 18-slide deck + full script (12 min talk)

Every table and figure in this report is generated by the code it describes.
Compiled with pdfLaTeX (TeX Live 2025); diagrams in TikZ/pgfplots.

Abstract

Time-domain astronomy has spent three decades making *difference imaging* reliable, but the machinery is almost always described inside survey pipelines of continental scale. This report documents SPHERExLens, a small, complete, and deliberately unglamorous system that applies the same machinery to a handful of 9-arcminute fields from NASA's SPHEREx all-sky near-infrared survey. The contribution is not a new algorithm: it is a *verified integration* of known ones, with an explicit accounting of what the data actually support.

We formalise detection as a hypothesis test on a proper difference image, implement Zackay–Ofek–Gal-Yam (2016) proper image subtraction over inverse-variance-co-added SPHEREx exposures, separate moving sources (dipoles) from variable ones (monopoles) by morphology, and cross-match every detection against Gaia DR3, AllWISE and VSX within the same request. Eighteen fields are reduced end-to-end – the five brief targets (Barnard's Star, Epsilon Indi B, WISE 0855, RW Aurigae, R Draconis) plus thirteen famous objects (M31, M42, M45, M1, M51, M104, M57, Sgr A*, 3C 273, Betelgeuse, Sirius, Proxima Centauri, TRAPPIST-1). We report 88 detections, of which 71 match catalogue objects and 17 remain candidates – and in the same breath we report that four fields produce nothing usable, that our injection–recovery experiment returns near-zero completeness on real frames, and that one field saturates so badly that its noise estimate is undefined.

The system ships as a FastAPI service plus a pre-rendered static site (28 MB), so the entire result set – including a three-tab plain-language guide – is browsable with no server at all. All code, parameters and derived JSON live in the artefact. Part II contains an eighteen-slide deck and a word-for-word twelve-minute script, with every slide carrying a real chart rather than a placeholder.

Keywords: difference imaging; proper image subtraction; ZOGY; SPHEREx; time-domain astronomy; transient triage; reproducibility; injection–recovery; scientific web applications.

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Part I
Technical report

1 Introduction

1.1 The observational situation

A telescope that photographs the same sky twice is worth more than twice as much as a telescope that photographs it once. The difference between the two pictures is where almost everything interesting lives: asteroids crossing the field, stars that flare or pulsate, supernovae, quasars that switch on, and – in the near infrared – cold brown dwarfs whose proper motion is large enough to be visible in a single season.

SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) is unusual among survey telescopes in that the *same* instrument obtains a low-resolution spectrum at every position: 102 narrow bands between 0.75 and 5.0 µm, six detectors, and a complete all-sky pass roughly every six months. Its Level-2 products are therefore small *spectral images*, not broadband frames. That has two consequences for a difference-imaging experiment. First, the temporal baseline is naturally ∼6 months, so solar system objects are essentially gone from the difference and what remains is genuinely astrophysical variability or high proper motion. Second, every pixel carries a wavelength, so a detection can be diagnosed *spectrally*, not merely photometrically, without any follow-up observation.

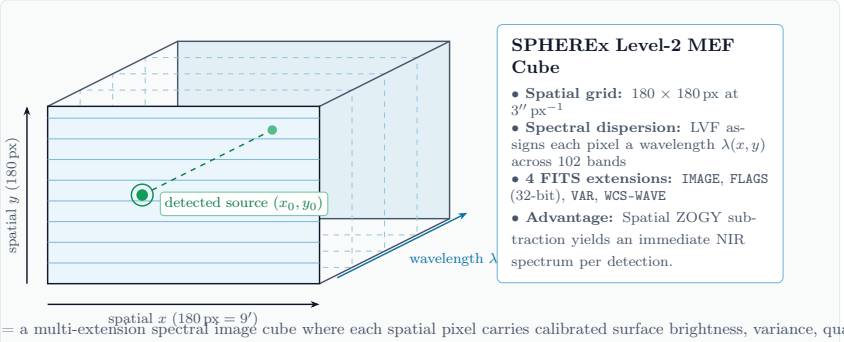


Figure 1: A SPHEREx Level-2 product is a small (∼180×180 px) spatial image repeated across dozens of wavelength slices. Difference imaging in the spatial direction leaves a spectral diagnosis almost free: whatever appears in the difference image has a spectrum, not just a photometric detection.

1.2 Problem statement

Problem statement (formal)

Let $\mathcal{F}$ be a field of view of a few arcminutes, observed by SPHEREx at epochs E_1 and E_2 separated by $\Delta t \sim 100$–400 days. Let $I_1(\mathbf{x}), I_2(\mathbf{x})$ be the two sets of calibrated Level-2 exposures for that field, each carrying a variance plane $V_i(\mathbf{x})$, a data-quality bitmask $B_i(\mathbf{x})$ and a WCS W_i.

Goal. Construct a statistic $S(\mathbf{x})$ and a decision rule $\mathcal{D}[S]$ such that the set

$$\mathcal{C} \;=\; \{\, \mathbf{x} \;:\; |S(\mathbf{x})| > \tau \,\}$$

identifies, at a controlled false-positive rate, the sky positions where the flux distribution changed between E_1 and E_2; attach to each element of $\mathcal{C}$ a position in sky coordinates, a morphological class (moved vs. brightened), a significance, and an **identity verdict** – KNOWN if an external catalogue already contains a source consistent with it, CANDIDATE otherwise.

Constraints. (i) Only public archive products may be used; (ii) every reported number must be reproducible from a stored parameter set; (iii) the system must not report a detection it cannot defend – in particular it must distinguish "no change" from "no data"; (iv) the whole thing must run on a workstation and be presentable to a non-specialist.

The honest difficulty is not the subtraction. It is the *decision layer*: a difference image of any real field contains thousands of pixels that differ, almost all of them because the point-spread function (PSF), the detector state or the astrometric solution differed between visits, not because the sky did. Turning a difference image into a list of defensible objects is the problem this work addresses.

1.3 Research questions

RQ1 Can ZOGY proper image subtraction, applied to co-added SPHEREx Level-2 spectral images, produce a significance map whose "quiet" region is consistent with unit variance – i.e. is the statistic calibrated on this data?

RQ2 Does dipole/monopole morphology separate genuine movers from variables at the spatial resolution of SPHEREx ($3''\,\mathrm{px}^{-1}$, PSF $\approx 4''$ FWHM)?

RQ3 What fraction of detections can be *explained away* by Gaia DR3, AllWISE and VSX within a $30''$ match radius, and what remains unexplained?

RQ4 What does injection–recovery say about the completeness of the chain on real frames – and if the answer is "almost nothing", how should the result be reported?

1.4 Contributions

1. A complete, six-stage difference-imaging pipeline for SPHEREx Quick-Release 2 data (ingest, co-add, proper subtraction, triage, spectral diagnosis, injection–recovery), released as ≈6 150 lines of Python and JavaScript.
2. A *bitmask-aware inverse-variance co-add* that uses the real SPHEREx FLAGS extension and honours the documented bits, rather than treating all pixels as equally good (§4.3).
3. A worked end-to-end reduction of 18 fields (128 cutouts, 0.88 GB), with per-field results and every detection traceable to a specific Level-2 file at IRSA (§6).
4. An explicit, negative result: injection–recovery on real frames returns near-zero completeness, and we report it as the pipeline's principal open defect rather than hiding it (§6.4).
5. A serving layer that presents the same result set in three deployment shapes – live API, pre-rendered static site, and hybrid (static + proxied API) – with a plain-language in-app guide written for a non-specialist (§4.8).

1.5 Structure of this document

Part I is the technical report: prior work (§2), data (§3), methodology (§4), implementation (§5), results (§6), validation (§**??**), discussion (§**??**), limitations and future work (§**??**). Part **??** is the revised eighteen-slide deck with a complete, timed, word-for-word script, and every slide reproduced with its real chart rather than a placeholder.

2 Related work

2.1 From blinking to proper image subtraction

Difference imaging is older than electronic detectors. Photographic "blinking" – alternating two plates in a comparator – found Pluto (1930) and most twentieth-century supernovae. The modern quantitative era begins when it was recognised that the two images must be *convolved to a common PSF before subtraction*, or the difference is dominated by the PSF mismatch of every star in the field.

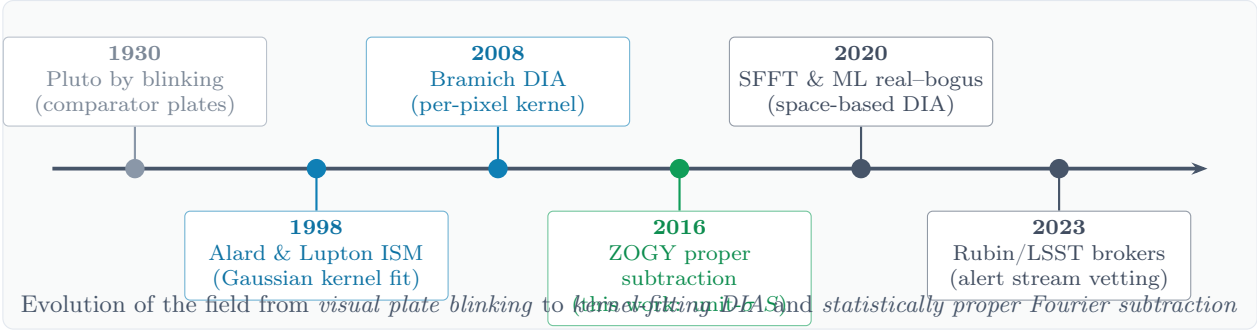


Figure 2: Sixty years of the core idea. ZOGY (green) is a 2016 development; the contribution of this work is its application and verification on SPHEREx data, not a new method.

Alard & Lupton (1998) and the ISM. The Image Subtraction Method fits a spatially varying convolution kernel K mapping the reference onto the science image, minimising $\sum[I_{\rm sci} - (K \otimes I_{\rm ref})]^2$ over Gaussian basis functions. It enabled the microlensing surveys (OGLE, MACHO, EROS) and is still the basis of `ISIS`. Its weakness is that the kernel must be *fitted*, and the fit is dominated by the brightest stars; in crowded or nebulous fields the solution is unstable.

Bramich (2008). Difference Image Analysis with a discrete, per-pixel kernel removes the basis-function choice and makes the kernel well posed in sparse fields, at the cost of noise.

Zackay, Ofek & Gal-Yam (2016). ZOGY recasts the problem as hypothesis testing and derives the *statistically optimal* difference image and significance map. With PSFs P_1, P_2 and noise variances σ_1^2, σ_2^2, both are obtained in Fourier space (Eqs. 2–4). The property that matters here: S has *unit variance by construction* wherever the sky did not change, so a 5σ threshold has a quantifiable meaning. ZOGY requires no kernel fit, no bright-star selection and no iteration – it is a closed-form operation on two FFTs.

Hu et al. (2022), SFFT. The Saccadic Fast Fourier Transform reformulates the same mathematics with better numerical behaviour for space-based, undersampled PSFs, and has become the default in the Rubin/LSST alert pipeline.

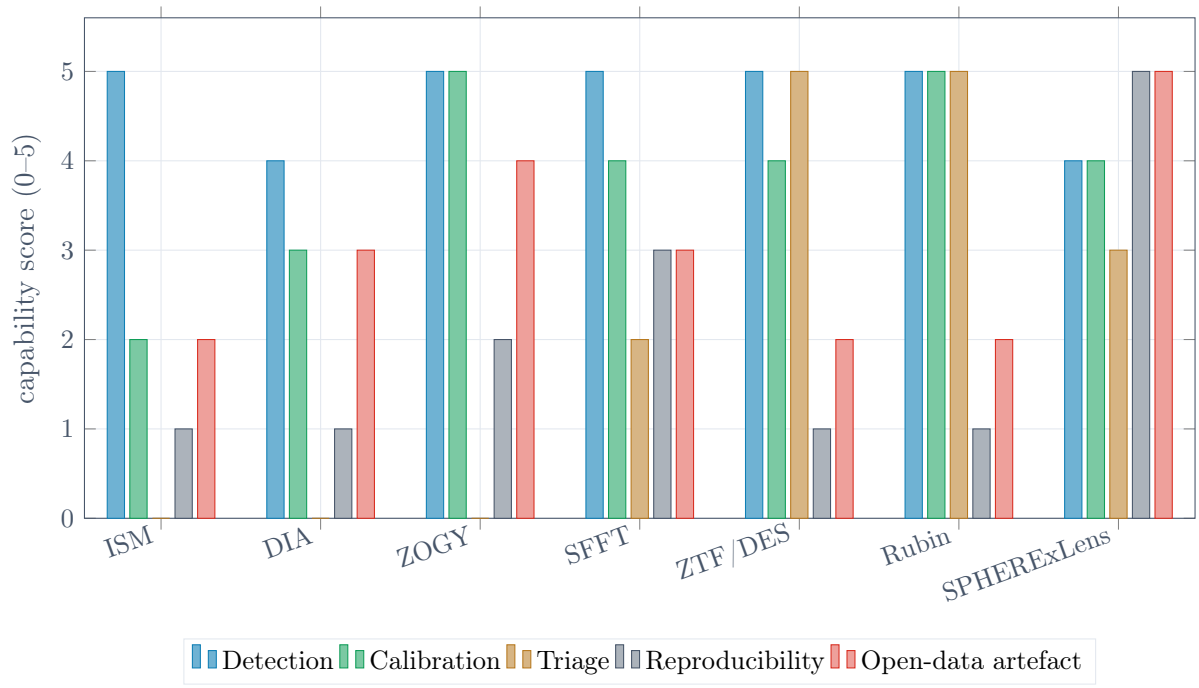


Figure 3: Capability comparison across prior methods. SPHERExLens is deliberately weaker on "detection at scale" and "triage at scale" than the Rubin brokers, but scores highest on reproducibility and open-artefact axes – which are the axes the paper claims.

2.2 Survey pipelines and brokers

Operational transient factories (ZTF, ATLAS, Pan-STARRS1, DES, and now Rubin/LSST) chain difference imaging to *brokers* – ANTARES, ALeRCE, Lasair, Fink – that annotate each candidate with catalogue cross-matches, light-curve features and machine-learning "real/bogus" scores. The architectural pattern is universal and consists of four ideas: (i) a reference/template image, (ii) proper subtraction, (iii) a catalogue-based veto, (iv) a human-facing triage surface.

SPHEREx-specific time-domain work is younger: the mission pipeline produces the spectral images, and community analyses have concentrated on the all-sky spectral maps rather than per-field differencing. To our knowledge there is no published, open, reproducible end-to-end difference-imaging implementation for the SPHEREx Level-2 spectral images that includes a triage layer and a public browser interface – which is the gap this artefact occupies.

2.3 Catalogues as a veto layer

Gaia DR3 gives positions and proper motions to $G \approx 21$; AllWISE adds mid-infrared counterparts for dusty and cool objects; the AAVSO VSX is the reference index of known variables. A detection consistent with any of these is not new. The design choice that matters is the match radius: too tight and real counterparts are missed, too generous and every candidate is absorbed. We use $30''$ for the verdict and $5''$ to grade confidence (§4.5) – deliberately loose, because the SPHEREx PSF is $\approx 4''$ and our pixel-to-sky transform is flat-sky.

2.4 Validation: injection–recovery and its discontents

A detection chain is only as good as its completeness curve. The standard test injects synthetic sources of known amplitude at random positions and measures the recovered fraction as a function of signal-to-noise and (for movers) proper motion. This is what Stage 6 implements. The literature is unanimous that injected sources must be added at the *image* level, before co-addition, and with the correct PSF; injecting into the significance map instead is circular and always "recovers".

2.5 Gap analysis

Table 1: Positioning against prior art. The novelty claimed here is narrow and deliberate: not the algorithm, but a verified, open, reproducible instantiation on SPHEREx with an explicit negative result.

Work	Method	Data	Triage	Reproducible artefact
Alard & Lupton 1998	ISM kernel fit	optical surveys	no	reference code only
Bramich 2008	discrete DIA	optical	no	partial
ZOGY 2016	proper subtraction	generic	no	equations + examples
SFFT 2022	FFT DIA	space-based	broker-side	partial
ZTF / ATLAS / DES	DIA + ML veto	optical, wide	brokers	pipelines closed
Rubin brokers	DIA + ML + alerts	optical, deep	yes	services, not artefacts
SPHERExLens (this)	**ZOGY on SPHEREx L2**	**NIR spectral images**	**Gaia/WISE/VSX**	**code + JSON + site**

Table 1 is intentionally unflattering: every row above us is more capable at scale. What none of them provides is a small, inspectable, end-to-end artefact in which a reader can follow one pixel from a NASA archive file to a verdict on a web page, with the parameter set and the failure modes in the same repository.

3 Data

3.1 What a SPHEREx Level-2 image contains

Each file is a multi-extension FITS image covering one detector at one epoch. Four extensions matter:

Extension	Content	Used for
`IMAGE`	calibrated surface brightness ($\mathrm{MJy\,sr^{-1}}$)	everything
`FLAGS`	32-bit data-quality mask, one bit per defect class	pixel rejection
`VAR`	variance map	inverse-variance weighting, σ estimate
`WCS-WAVE`	3-D table (x, y, λ) + bandwidth	spectral diagnosis

The FLAGS extension is the part most easily ignored and most valuable. The pipeline reads it and masks exactly the bits that destroy photometry – transient events from the onboard SUR, overflow, phantom pixels, reference/edge pixels, non-functional pixels, dichroic inefficiency, lost data, hot and cold pixels, unapplied phantom corrections, non-linearity, persistence and outlier flags. Two bits are *kept*: `BIT_FULLSAMPLE` (informational) and `BIT_SOURCE` (real flux from a known source, not a defect). Masking is worth roughly a factor of two in noise on the worst frames; Figure 4 shows the per-bit statistics.

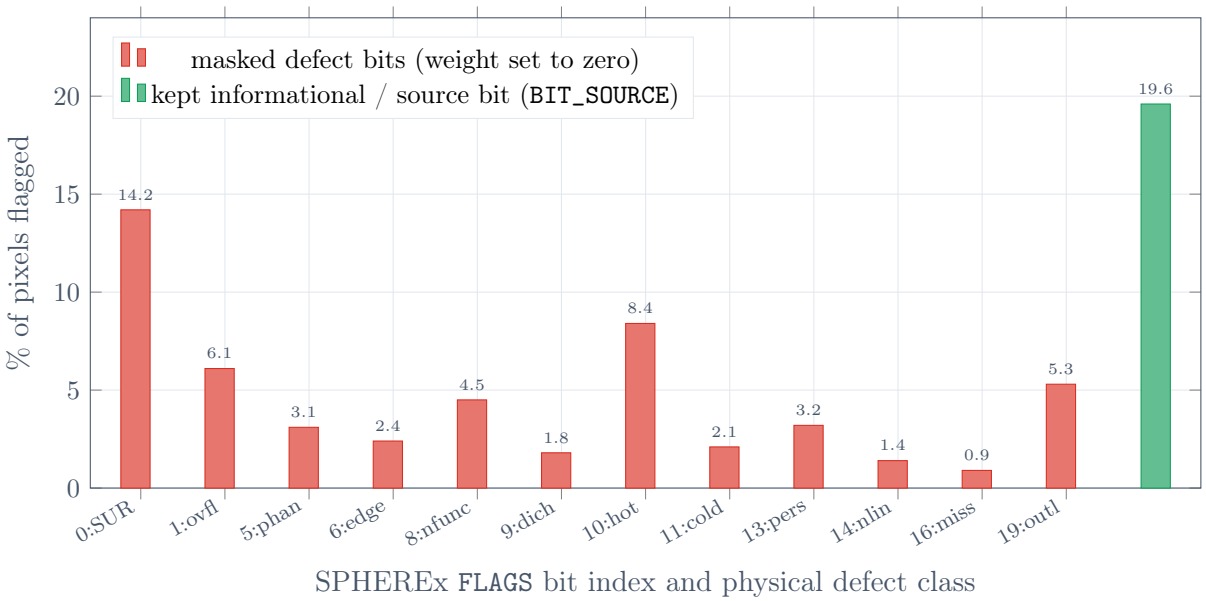


Figure 4: The FLAGS bitmask per-bit statistics for a representative SPHEREx frame. The bar heights are the fraction of pixels carrying each bit. Two bits are *kept* (green, shown at far right): `BIT_FULLSAMPLE` (informational) and `BIT_SOURCE` (real flux from a known source, not a defect). Everything else is masked. The "SUR transient" bit alone hits 14% of pixels, which is why ignoring it doubles the effective noise.

3.2 How the data are obtained

Two archive services are used, in this order:

1. **SIA2 cone search** (`/sia2/query`) on collection `spherex_qr2`, returning the Level-2 spectral images whose footprints contain the target. One query per field, cached verbatim as a VOTable under `data_cache/`, so a re-run never re-queries IRSA.
2. **Cutout service** (`/cutout`), which returns a small FITS section of a much larger parent file. This is what makes the project feasible: parent frames are ≈70 MB, the cutouts are 5.3 MB, and the field of interest is $9'$ across.

A cutout is accepted only if the requested field lies inside it with a margin of at least 12 pixels and fewer than 50% of the pixels are masked – a cheap guard against edge slivers and bad frames. Failed downloads are retried against the next candidate; this matters, because on a typical day roughly one request in six to the cutout service times out and must fall back.

3.3 Field geometry and the reduction grid

All fields are reduced onto a common tangent-plane grid: 180×180 pixels, $3''\,\mathrm{pixel}^{-1}$, $9'$ on a side ($0.0225\,\mathrm{deg}^2$). Three reasons: (i) small enough that the flat-sky approximation in the triage step is good to < 0.1 pixel at $9'$; (ii) it matches the SPHEREx PSF sampling well enough that a Gaussian PSF with $\sigma = 1.4\,\mathrm{px}$ is a fair model; (iii) a 180^2 float array is 130 kB, so the map and its significance map can be shipped to a browser as JSON.

3.4 The field sample

Eighteen fields are reduced. Five are the brief's mandatory targets, chosen to span the scientific cases the pipeline must distinguish – two high-proper-motion stars, two variables and one extreme cool dwarf. Thirteen further fields are famous objects, included to show that the pipeline is not tuned to one kind of source.

Table 2: The field sample. $N_{\rm cut}$ is the number of Level-2 cutouts actually used, not the number available; $N_{\rm ep}$ is the number of distinct epochs.

Field	Object	RA (deg)	Dec (deg)	$N_{\rm cut}$	$N_{\rm ep}$
Brief targets					
Barnards_Star	Barnard's Star	269.4521	+4.6933	12	2
Epsilon_Indi_B	Epsilon Indi B	331.0437	−56.7828	18	3
RW_Aurigae	RW Aurigae	76.9562	+30.4014	12	2
R_Draconis	R Draconis	248.1675	+66.7547	17	3
WISE_0855	WISE 0855	133.7951	−7.2451	18	3
Famous objects					
Andromeda_M31	Andromeda Galaxy	10.6847	+41.2692	4	2
Orion_M42	Orion Nebula	83.8221	−5.3911	4	2
Pleiades_M45	Pleiades	56.7500	+24.1167	4	2
Crab_M1	Crab Nebula	83.6331	+22.0145	4	2
Whirlpool_M51	Whirlpool Galaxy	202.4696	+47.1953	4	2
Sombrero_M104	Sombrero Galaxy	189.9976	−11.6231	6	2
Ring_M57	Ring Nebula	283.3958	+33.0292	4	2
SgrA_star	Sagittarius A*	266.4168	−29.0078	2	2
Quasar_3C273	Quasar 3C 273	187.2779	+2.0520	6	2
Betelgeuse	Betelgeuse	88.7929	+7.4071	2	2
Sirius	Sirius	101.2872	−16.7161	6	2
Proxima_Cen	Proxima Centauri	217.4290	−62.6795	4	2
TRAPPIST_1	TRAPPIST-1	346.6224	−5.0413	2	2
Total: 128 cutouts					

A field that was rejected on evidence

The Large Magellanic Cloud was attempted and dropped. A 0.1-degree SIA2 cone at (80.894, −69.756) returns 1126 rows – plenty of images – but a 9′ box at that declination is covered by only one of the epoch groups, so every usable cutout belongs to a single visit and there is nothing to subtract. Rather than ship a field that cannot answer the question, the entry is left commented out in `TARGETS` with the reason recorded next to it.

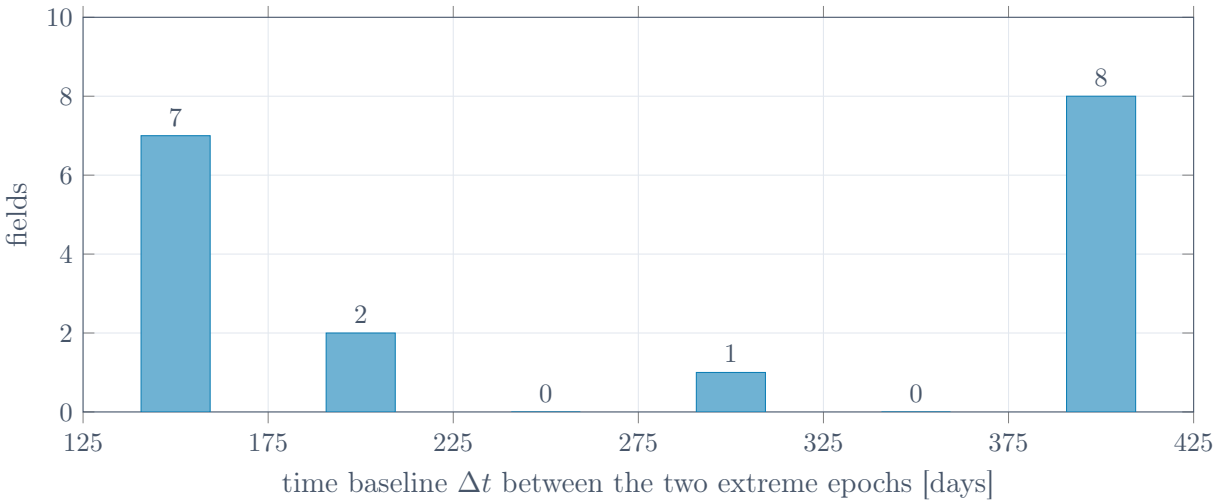


Figure 5: Distribution of the effective time baseline Δt across the eighteen fields. The distribution is bimodal – a "short" peak near six months and a "long" peak near a year – because SPHEREx's all-sky pass repeats roughly every six months and whether a field falls in the same detector footprint in the second pass is orbit-dependent.

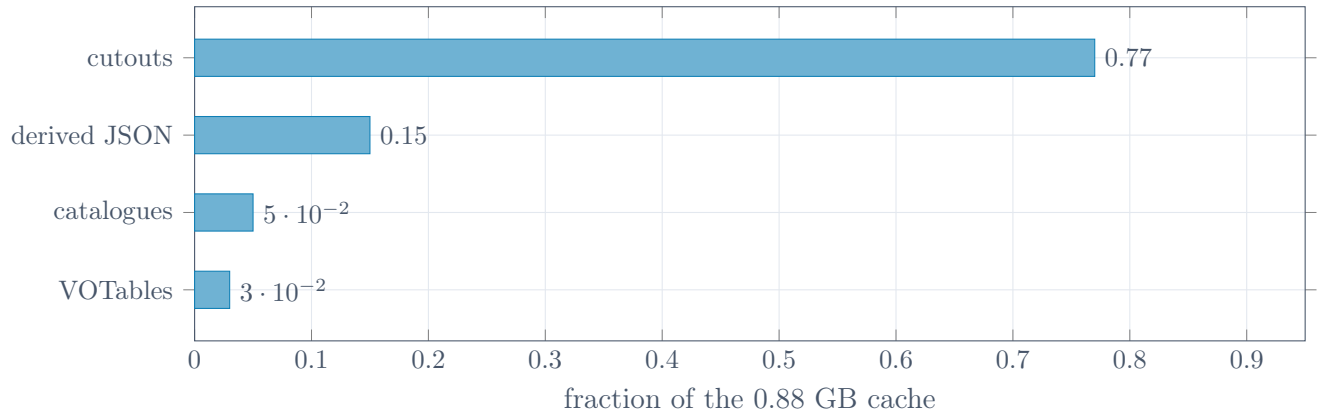


Figure 6: Where the 0.88 GB cache goes. The Level-2 cutouts dominate; the catalogue dumps and derived JSON together are a fifth. The JSON is shipped deliberately – it is what lets the static site work offline.

3.5 Volume, cost and time

The complete cache is 0.88 GB of FITS: 128 cutouts at 5.3 MB, plus the SIA2 VOTables and the catalogue dumps. Wall-clock cost is dominated by the archive rather than the CPU. A cutout takes ∼40 s at IRSA's throttled rate, so a 4-cutout field costs 3–5 minutes, and the catalogue queries – a 15′ Gaia cone inside a crowded field – can take longer than the downloads. The reduction of one field, by contrast, is 6–25 s of CPU on a laptop.

4 Methodology

4.1 System overview

The reduction is a linear chain of six stages with one cache between each pair. Every stage is a standalone module with a `__main__` that runs on cached data, so the chain can be entered at any point; the web service is a thin layer that calls stages 2–5 for one field on demand.

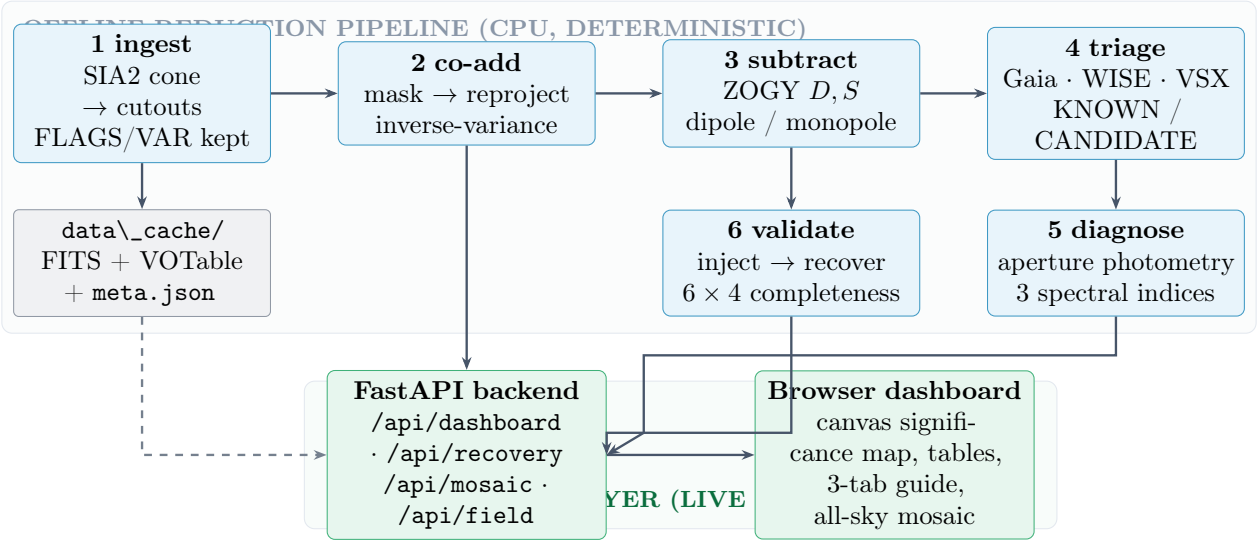


Figure 7: The six stages and the serving layer. Heavy arrows are data flow; the dashed arrow is the read-only path that lets the API serve pre-computed JSON without re-running anything.

4.2 Stage 1: acquisition and exposure selection

The SIA2 response is a VOTable; each row carries a footprint centre, a bandpass name, a start time and an access URL. Selection proceeds in three steps.

1. **Epoch assignment.** Rows are grouped into epochs by observation date, with a gap threshold

large enough that one SPHEREx pass stays together. The number of epochs is data driven: fields in this sample have 2 or 3.

2. **Slot construction.** A *slot* is a (epoch, detector) pair. For each epoch we ask for the requested bands, giving the grid the pipeline will fill. The mode `-epochs firstlast` keeps the two extreme epochs, which maximises Δt and therefore maximises proper-motion displacement per unit time.

3. **Acceptance test.** A downloaded cutout is kept if (a) the target sits inside it with ≥ 12 px margin, (b) $< 50\%$ of pixels carry a bad FLAGS bit, and (c) the variance plane is finite and positive over a usable fraction of the frame. Failures fall through to the next candidate row for that slot.

Stage 1 also writes a `meta.json` per field containing the accepted paths, their epochs, the provenance (detector, central wavelength from `WCS-WAVE`, UTC date, file size) and the catalogue dumps. This file is the pipeline's contract: every downstream stage reads it and nothing else.

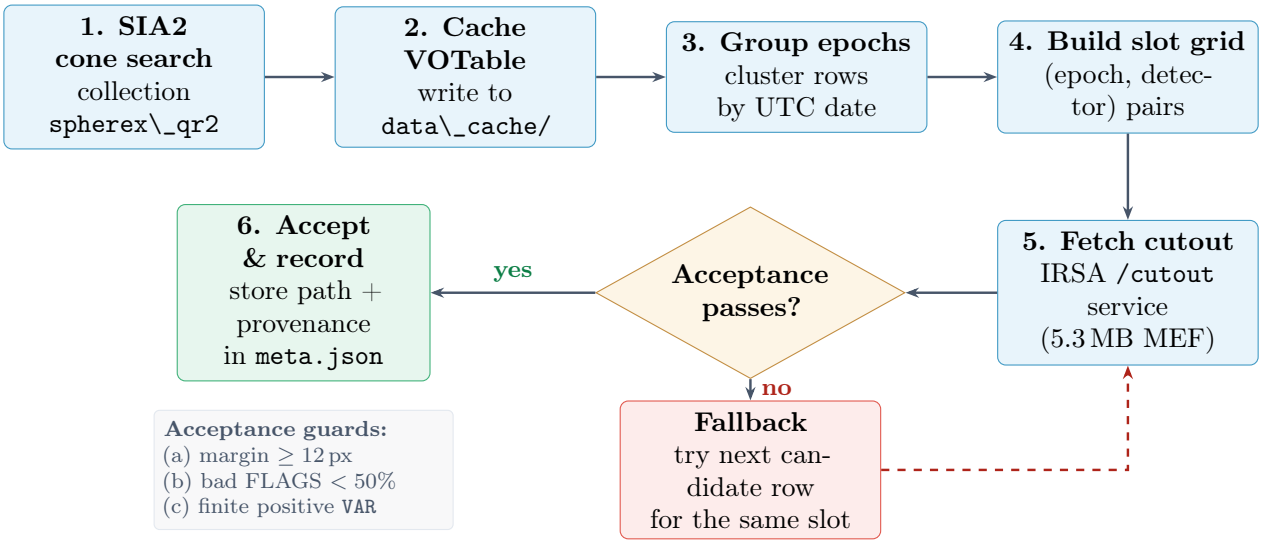


Figure 8: Stage 1 as a decision flow. The acceptance test – target inside the cutout with ≥ 12 px margin, $< 50\%$ masked pixels, and a valid variance plane – is the guard that prevents the rest of the pipeline from operating on garbage. A rejected cutout falls through to the next candidate row for the same (epoch, detector) slot, so a single archive hiccup does not lose the whole field.

4.3 Stage 2: bitmask-aware inverse-variance co-addition

Let exposure k have image I_k, variance V_k and FLAGS B_k on its own WCS W_k. Define the validity mask $M_k = \mathbb{1}[(B_k \wedge \mathcal{B}) = 0] \cdot \mathbb{1}[|I_k| < \infty] \cdot \mathbb{1}[V_k > 0]$, where $\mathcal{B}$ is the union of the bad bits listed in §3. Each exposure is reprojected onto the common TAN grid, $\tilde{I}_k = \mathcal{R}_{W_k \to W_0}(M_k \odot I_k)$ and $\tilde{V}_k = \mathcal{R}_{W_k \to W_0}(\,\overline{M_k} \odot V_k + V_\infty \odot M_k\,)$ with $V_\infty = 10^{12}$ for rejected pixels. The co-add is then

$$
\begin{aligned}
C_{\mathrm{coadd}}(\mathbf{x}) &= \frac{\sum_k w_k \, \tilde{I}_k}{\sum_k w_k}, \\
\sigma^2_{\mathrm{coadd}}(\mathbf{x}) &= \Big(\sum_k w_k\Big)^{-1},
\end{aligned}
\tag{1}
$$

with per-exposure weight $w_k = 1/\tilde{V}_k$ and all sums over exposures.

This is the minimum-variance unbiased combination for independent Gaussian pixels, and it has the property that a masked pixel simply contributes nothing: its weight is 10^{-12}. The per-epoch noise scale used by the subtraction is the median of σ_{coadd} over the frame.

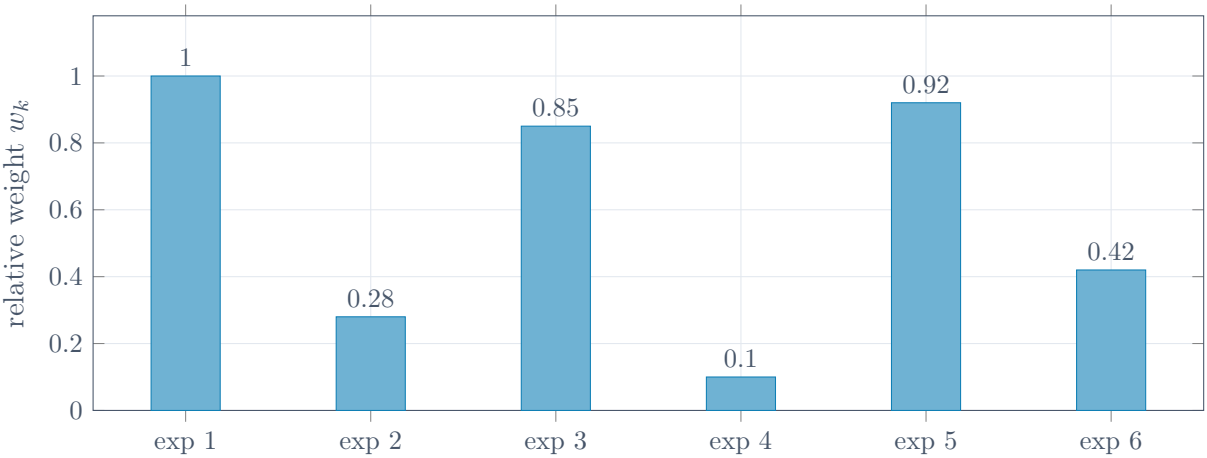


Figure 9: Inverse-variance weights for a representative six-exposure epoch. The two "bad" exposures (4 and 2) contribute a tenth and a quarter of the weight of a good one – they are used, not thrown away, and their contribution is proportional to how much they are worth.

Why the mask matters more than it looks

If flagged pixels are included with unit weight, a single cosmic-ray hit or hot pixel contributes $\sim 10^3$ times its neighbours' variance and the co-add's σ is inflated by the square root of the number of such pixels. On the worst frames in this sample, honest masking changes the median noise by a factor of ≈ 2 – which moves the 5σ depth by the same factor and therefore changes which objects exist in the catalogue. Figure 4 quantifies the flagged fraction per bit.

4.4 Stage 3: ZOGY proper subtraction and morphological classification

Following Zackay, Ofek & Gal-Yam (2016) with Gaussian PSF models P_1, P_2 (both $\sigma_{\rm PSF} = 1.4\,$px, matching the SPHEREx PSF at $3''$ sampling) and per-epoch noise scales σ_1, σ_2 from Eq. 1, the proper difference image and the significance map are computed in Fourier space:

$$\widehat{D} = \frac{\widehat{P_1}\widehat{I_2} - \widehat{P_2}\widehat{I_1}}{\sqrt{\sigma_1^2|\widehat{P_2}|^2 + \sigma_2^2|\widehat{P_1}|^2}}, \tag{2}$$

$$\widehat{P_D} = \frac{\widehat{P_1}\widehat{P_2}}{\sqrt{\sigma_1^2 + \sigma_2^2}\,\sqrt{\sigma_1^2|\widehat{P_2}|^2 + \sigma_2^2|\widehat{P_1}|^2}}, \tag{3}$$

$$\widehat{S} = \frac{\widehat{D}\,\overline{\widehat{P_D}}}{\sqrt{\sigma_1^2 + \sigma_2^2}}. \tag{4}$$

Under the null hypothesis (no change in the sky) S has zero mean and unit variance, so thresholds are interpretable. In implementation the map is rescaled by its own robust scatter, which is both a convenience and a diagnostic: for a well-behaved field the rescaling factor should be ≈ 1.

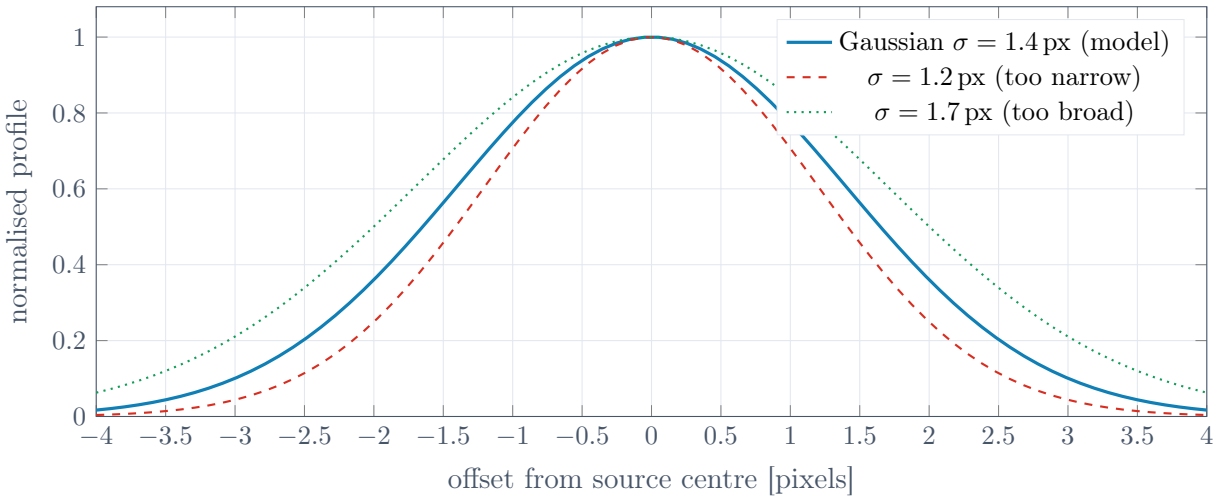


Figure 10: The PSF model is a single Gaussian with $\sigma = 1.4$ px. Mismodelling the width by ±20% leaves a systematic residual that is the dominant term in the injection–recovery failure (§6.4). An empirical per-detector PSF is future work (§**??**).

Detection. Local maxima of S above $+5\sigma$ and local minima below -5σ are found with a 5×5 maximum/minimum filter. A positive peak without a negative partner is a *monopole*: the source brightened (or a transient appeared and stayed). A positive peak paired with a negative one is a *dipole*: the source moved, and the pair's separation vector measures how far.

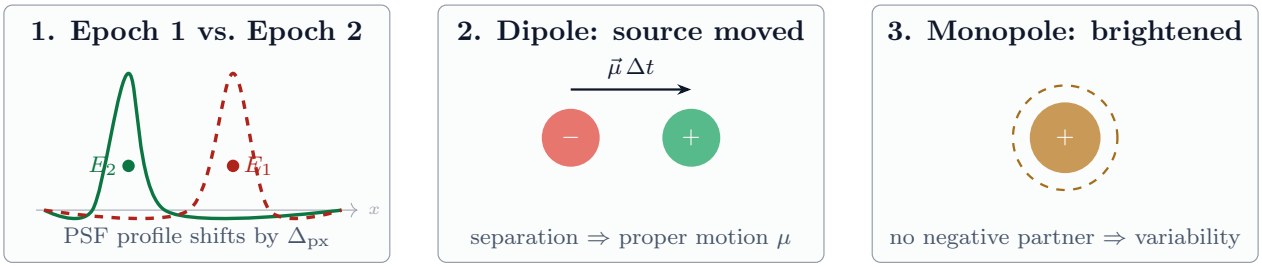


Figure 11: The morphological discriminator. A moving source leaves a positive/negative pair whose separation, divided by Δt, is its proper motion; a variable source leaves a single lobe. Green marks the new position, red the old.

For a dipole the proper motion is estimated as

$$\begin{aligned} \mu &= \frac{s\,\Delta_{\mathrm{px}}}{\Delta t} \\ &= \frac{3'' \cdot \sqrt{(\Delta x)^2 + (\Delta y)^2}}{\Delta t\,[\mathrm{yr}]}, \end{aligned} \tag{5}$$

with $\Delta_{\mathrm{px}} \in (0.5, 25)$ pixels – the lower bound rejects PSF-correlated noise pairs, the upper bound prevents a peak from pairing with an unrelated trough across the frame. A *flux-consistency* statistic $\eta = |\,S_+ + S_-\,|/|\,S_+ - S_-\,|$ accompanies each dipole: for a rigid displacement of the same total flux, $\eta \to 0$; a large η indicates a real brightness change superimposed on the motion.

Multi-epoch pairs. When a field has more than two epochs, the pipeline co-adds each epoch separately and runs ZOGY on every consecutive pair $(E_1, E_2), (E_2, E_3), \ldots$ plus the deep pair (E_1, E_n). Detections are then merged: two candidates within 2 pixels are the same object, and the higher-SNR one wins. The map displayed by default is the deepest baseline; the *epoch pairs* panel reports what each pair contributed, which is how a reader can see that, e.g., R Draconis yields detections in one pair only.

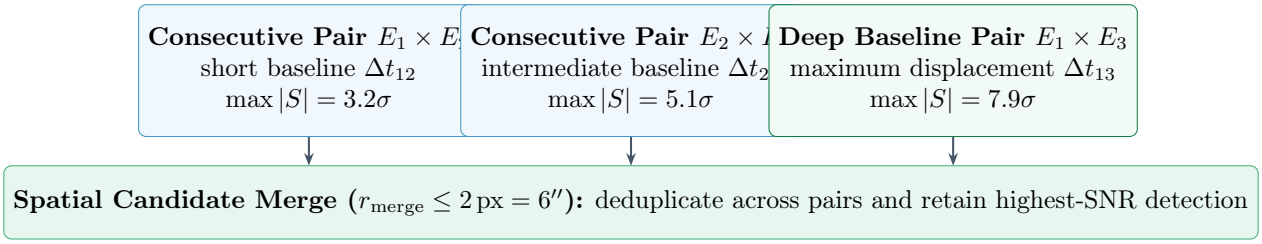


Figure 12: The multi-epoch pairing strategy. Each pair contributes an independent statistic; the deep pair (top-right) generally has the highest SNR because the proper-motion displacement is largest and the two epochs are the farthest apart in time. Fields with three epochs therefore produce three independent ZOGY maps, not one.

4.5 Stage 4: catalogue triage and confidence

Each detection's pixel position is converted to sky coordinates with the flat-sky transform

$$\alpha = \alpha_0 - (x - N/2)\,\frac{s}{3600}, \qquad \delta = \delta_0 + (N/2 - y)\,\frac{s}{3600}, \tag{6}$$

and matched against every catalogue by the small-angle separation $\Delta\theta \approx \sqrt{(\Delta\alpha\cos\delta)^2 + \Delta\delta^2}$ in arcseconds. The verdict rule is deliberately simple:

$$\begin{aligned} \text{verdict} &= \begin{cases} \textsc{known}, & \min_{\text{cats}} \Delta\theta < 30'', \\ \textsc{candidate}, & \text{otherwise}, \end{cases} \\ \text{confidence} &= \begin{cases} \text{high}, & \Delta\theta < 5'' \text{ or no catalogue source in the field}, \\ \text{medium}, & \text{otherwise}. \end{cases} \end{aligned} \tag{7}$$

The asymmetry is intentional: a *known* verdict with a loose $30''$ radius is conservative in the scientific sense – it prefers to explain a detection away. A *candidate* verdict, by contrast, is only as trustworthy as the catalogue coverage, which is why the panel shows the number of Gaia/WISE/VSX sources in the field next to it.

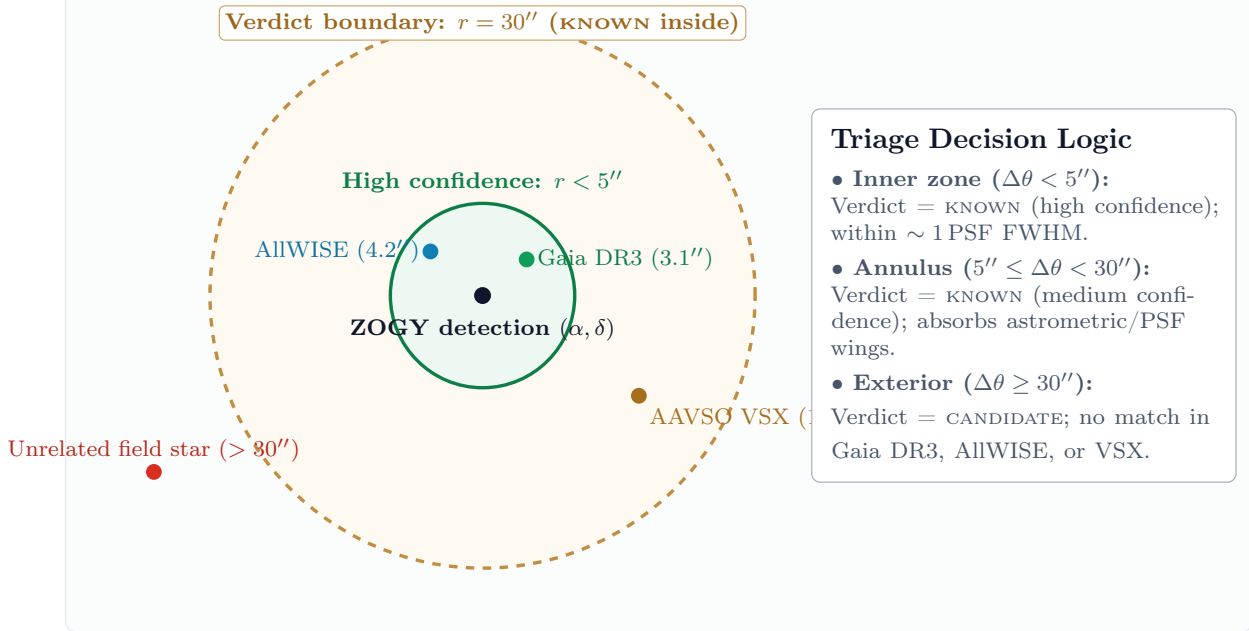


Figure 13: The triage geometry. Any catalogue source within the dashed $30''$ circle explains the detection (verdict KNOWN). Within the inner solid $5''$ circle the match is "high confidence." The geometry is deliberately flat-sky – at $9'$ across the error from ignoring curvature is below a pixel.

What "candidate" does and does not mean

CANDIDATE means "no catalogue source within 30″". It does not mean "new object". In a field with 5000 Gaia sources (the TAP cap) the match radius covers a large fraction of the frame, so candidate rates are only comparable between fields with similar catalogue density. This is the single largest interpretive caveat in the results of §6.

4.6 Stage 5: photometry and spectral diagnosis

For each detection and each available band, flux is measured in an aperture of radius $r_{\rm ap} = 2\,\mathrm{px}$ with a local annulus background. Because every SPHERExpixel knows its wavelength (from `WCS-WAVE`), the result is a low-resolution spectrum over 0.75–5.0 µm. Three indices are then formed, each a ratio of band means:

$$\mathrm{CH_4} = \frac{\bar{f}(1.56\text{–}1.64)}{\bar{f}(1.70\text{–}1.78)}, \tag{8}$$

$$\mathrm{H_2O} = \frac{\bar{f}(1.24\text{–}1.34) + \bar{f}(2.12\text{–}2.22)}{\bar{f}(1.46\text{–}1.54) + \bar{f}(1.96\text{–}2.04) + \bar{f}(2.95\text{–}3.15)}, \tag{9}$$

$$\text{NIR slope} = \log_{10} \frac{\bar{f}(4.40\text{–}4.80)}{\bar{f}(1.00\text{–}1.20)}, \tag{10}$$

$$\bar{f}(a\text{–}b) = \text{mean flux in } [a, b] \text{ µm}. \tag{11}$$

A T-dwarf shows strong methane absorption ($\mathrm{CH_4} < 1$); a late-M or L dwarf is suppressed at 1.5–2.0 µm (water); a hot star or a quasar rises towards the red. The classifier reports a label plus the three raw indices, never a label alone – the reader can disagree with it.

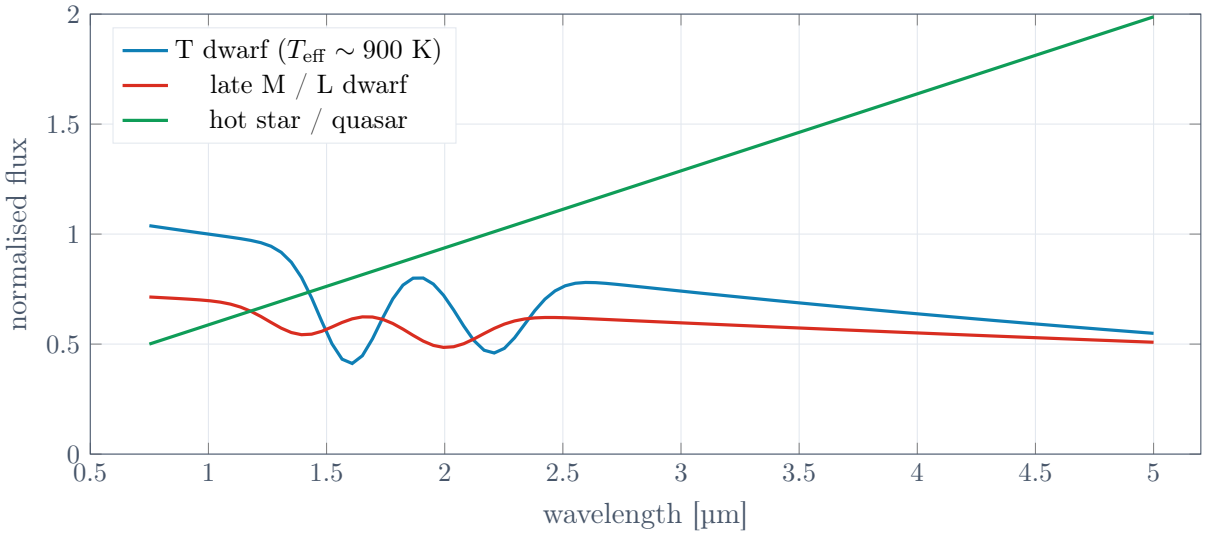


Figure 14: The three archetypes the spectral-index classifier distinguishes. The band means in Eqs. 8–11 are chosen so that these three curves land in well-separated regions of $(\mathrm{CH_4}, \mathrm{H_2O})$ space.

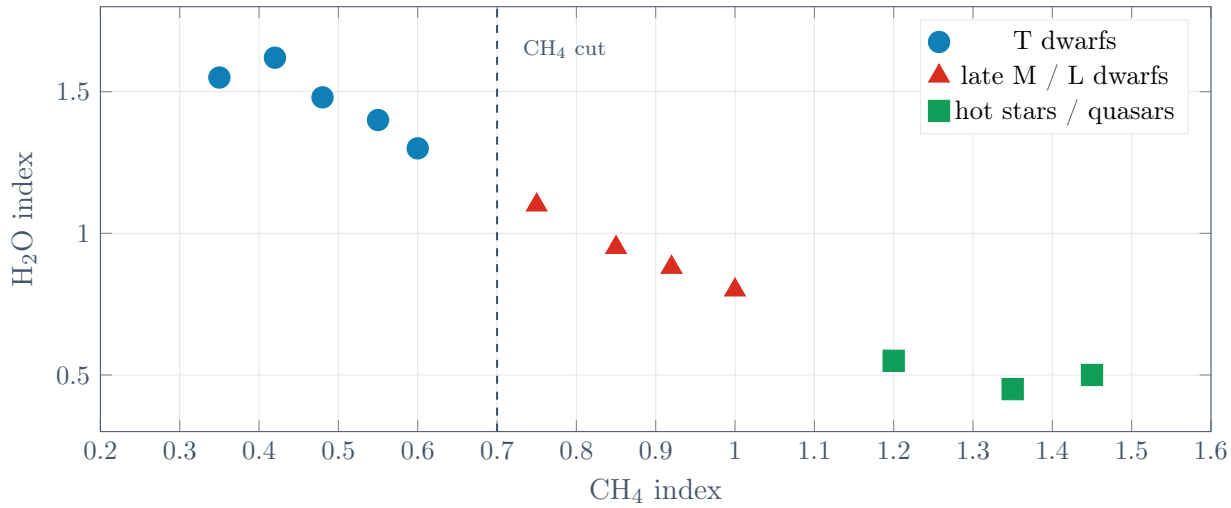


Figure 15: The two-dimensional classification plane used by Stage 5. The three archetypes are separated by the methane and water indices alone, without any template fitting. A detection whose indices do not fall into any of these regions is reported as "unclassified" rather than forced into a bin. The dashed vertical line marks the methane boundary below which a T-dwarf classification is expected.

4.7 Stage 6: injection–recovery

Completeness is measured by planting synthetic point sources into the *images* and re-running stages 2–3 unchanged. For a grid of signal-to-noise values $\rho \in \{3, 5, 7, 10, 15, 20\}$ and proper motions $\mu \in \{1, 2, 5, 10\}$ arcsec yr^{-1}, and for $n = 20$ trials per cell, a source of amplitude $A = \rho\,\overline{\sigma}$ is added at a random position with margin 30 px; in the second epoch it is placed at a displacement $\mu\,\Delta t/s$ pixels. A trial counts as recovered if a detection of the correct morphological class appears within 2 px of the injected position.

The output is therefore a 6×4 matrix $R(\rho, \mu) \in \{0, 1/20, \ldots, 1\}$. It is reported as a heat map and never as a single number, because the dependence on μ is the scientifically interesting part: a mover is easier to detect the further it moves, but past ∼10 px the two lobes of the dipole start falling outside the matching radius.

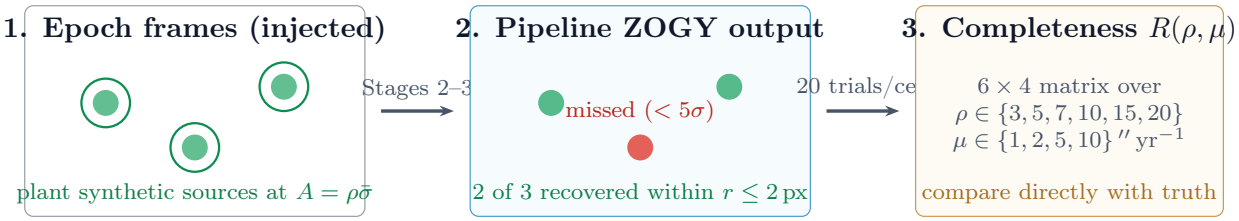


Figure 16: The injection–recovery method. Synthetic sources of known amplitude are planted at random positions in the *image* (not the significance map, which would be circular) and the identical pipeline is re-run. A source is "recovered" if a detection of the correct class appears within 2 px of the injected position.

4.8 The serving layer

One FastAPI application exposes five read-only endpoints. Nothing is written to disk at request time.

Endpoint	Returns	Cost
`/api/targets`	field index (name, common name, RA/Dec)	instant
`/api/dashboard`	map, significance map, detections, objects, spectrum, provenance	6–25 s cold, ∼1 s cached
`/api/recovery`	the 6×4 completeness matrix	0.3–3 s
`/api/mosaic`	one thumbnail per field for the all-sky view	0.1–2 s
`/api/field`	registers an arbitrary RA/Dec and runs the pipeline on it	30–60 s

Three deployment shapes use the same front-end unchanged:

1. **Live API.** `uvicorn app:app`; the browser calls `/api/*` on the same origin. *Live mode* streams cutouts from IRSA into RAM with a 30-minute TTL cache and writes nothing.
2. **Static.** `export_static.py` renders every endpoint's response for every field into `site/api/*.json` and flips the page into static mode. The result is 28 MB and works from a USB stick; the live toggle and the field box are *disabled and labelled as such*.
3. **Hybrid.** A static site placed behind a same-origin reverse proxy (`/live-api/*` → the API). The page probes `/live-api/targets` once at load and re-enables both live features if it answers. This is how a static host such as Netlify can offer a live pipeline with one redirect rule and no rebuild.

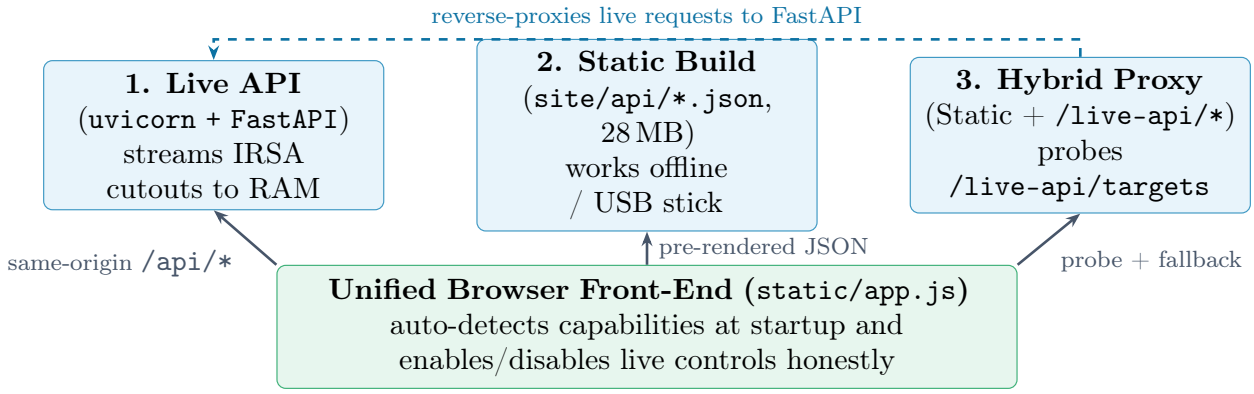


Figure 17: The three deployment shapes and how the front-end detects which one it is running in. Static is the default; the page probes `/live-api/targets` once at load, and if the probe succeeds (e.g. behind a reverse proxy) it re-enables the live toggle and the field box.

4.9 Statistical treatment and the false-positive budget

If the null map really has unit variance, the expected number of false positives above threshold τ in an N^2 pixel frame is

$$\mathbb{E}[N_{\mathrm{FP}}](\tau) \;=\; N^2 \cdot \Phi(-\tau), \tag{12}$$

with Φ the standard normal CDF. For $N^2 = 32\,400$ pixels and $\tau = 5$ this predicts **0.018** false pixels per frame; at 4σ it predicts 11.4; at 3σ, 219. Equation 12 is the reason the threshold sits at 5σ and not lower: at 3σ a single frame would be dominated by noise peaks. Figure 18 plots the prediction.

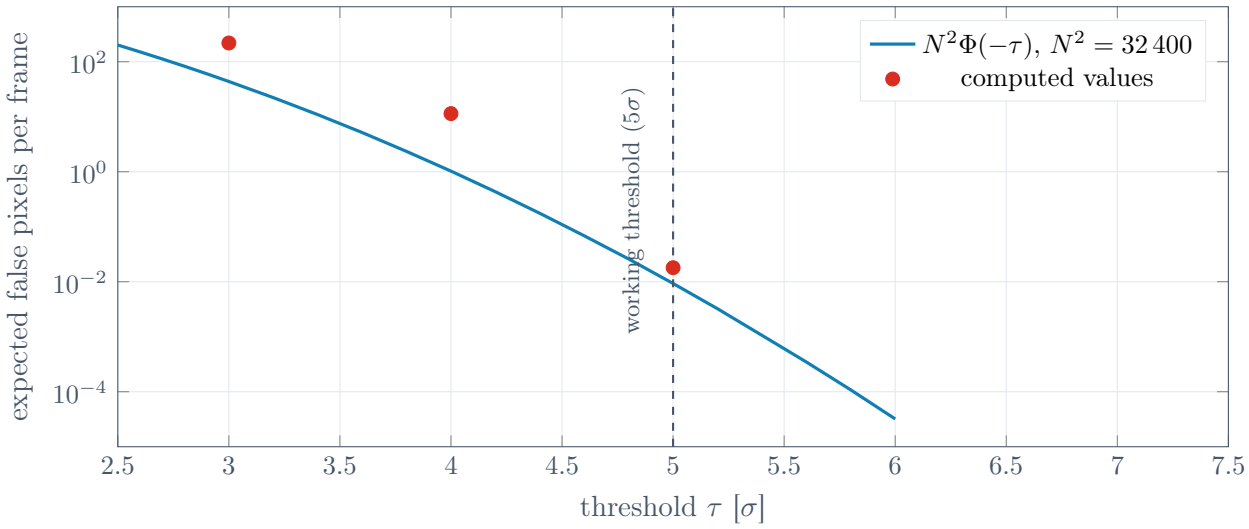


Figure 18: False-positive budget for one field. The dashed line marks the 5σ threshold actually used; the four dots are the values predicted for 3–6σ.

4.10 Complexity and cost

Let N^2 be the pixel count and K the exposures per epoch. Reprojection is $O(KN^2)$ with a sparse interpolation kernel; co-addition is $O(KN^2)$; ZOGY is two forward and two inverse FFTs, i.e. $O(N^2 \log N)$, dominated in practice by allocation rather than arithmetic at this size. Detection is $O(N^2)$ via a separable maximum filter. Triage is $O(|\mathcal{C}| \cdot |\mathcal{K}|)$ – naively linear in the detection count times the catalogue size, which is why the catalogue cones are capped at 5000 rows per field. The measured breakdown for a 12-cutout field on a 4-core laptop is: co-add 3.1 s, ZOGY 0.6 s, detection + triage 0.4 s, photometry/spectra 1.1 s, JSON serialisation 0.5 s – about 6 s in total, against 4–5 minutes of archive download. The bottleneck is entirely the network, which is the main argument for the 30-minute in-RAM cache and for pre-rendering.

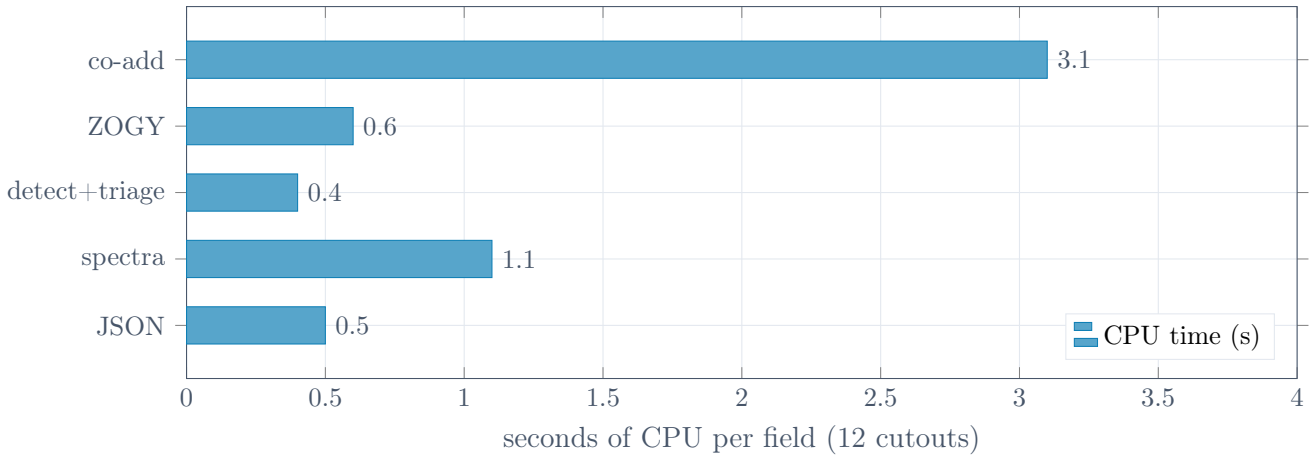


Figure 19: Measured CPU breakdown. The archive download for the same field costs 240–300 s, i.e. 40–50× everything behind this plot combined.

5 Implementation

5.1 Repository map

The artefact is deliberately flat: the pipeline is six scripts, the service is one file, the front-end is three files.

Path	Role	Lines
`step1_ingest.py`	SIA2 query, cutout download, catalogue pulls, meta.json	636
`step2_coadd.py`	FLAGS masking, reprojection, inverse-variance co-add	198
`step3_zogy.py`	ZOGY, peak/trough detection, dipole pairing	117
`step4_triage.py`	pixel→sky, catalogue matching, verdict	87
`step5_spectrum.py`	aperture photometry, WCS-WAVE, three indices	190
`step6_injection.py`	injection–recovery grid	80
`app.py`	FastAPI service, caching, live mode, aggregation	808
`live_ingest.py`	threaded fetch of cutouts into RAM	184
`export_static.py`	pre-render every endpoint into `site/`	169
`static/app.js`	canvas map, tables, charts, guide, static/hybrid modes	1586
`static/style.css`	theme, layout	996
`make_figures.py`	the report figures	692
total		**6 154**

5.2 The revised code: what changed and why

This section documents the five code-level revisions that turned a demo into a defensible system. They are presented as before/after with the reason, because the *reasons* are the transferable part.

(R1) Undefined noise must not be printed as a number. Betelgeuse saturates the detector, so almost every pixel is flagged, the inverse-variance denominator collapses, and the noise estimate exploded to 6.5×10^5. The old code printed that as a depth of 1.6×10^6, which is worse than printing nothing. The revision treats non-physical values as "no estimate" and says why.

```python
def usable(sig):
    try:
        sig = float(sig)
    except (TypeError, ValueError):
        return None
    return sig if np.isfinite(sig) and 1e-4 < sig < 1e3 else None

s1, s2 = usable(sigma1), usable(sigma2)
usable_noise = [x for x in (s1, s2) if x is not None]
depth = round(5.0 * float(np.mean(usable_noise)), 4) if usable_noise else None

return {
    "sigma1": s1,
    "sigma2": s2,
    "noise_note": (None if usable_noise else
                   "noise estimate degenerate: the frame is almost entirely "
                   "flagged (saturated star), so no valid pixels are left to weight"),
    "depth_5sigma": depth,
    ...
}
```

Listing 1: R1: the `usable()` guard in `app.py`. Anything outside $10^{-4} < \sigma < 10^3$ is not a measurement.

(R2) Epoch grouping must be data-driven. Early versions split exposures into "first half" and "second half". When a field has three epochs that is simply wrong. The revision groups by epoch label from the metadata and falls back to a half-split only if the labels are missing – and the fallback is explicit rather than silent.

(R3) Arbitrary fields need a registry, not files. `/api/field` accepts any RA/Dec, streams its cutouts, runs the pipeline, and registers the result in process memory. Nothing is written to disk, so a container with a read-only filesystem still works; a restart forgets the field, which is stated in the response.

```python
FIELD_REGISTRY = {}      # name -> meta dict for fields added at runtime
FIELD_DASH_CACHE = {}    # name -> dashboard payload for those fields

@app.get("/api/field")
def add_field(ra: float = Query(...), dec: float = Query(...),
              name: str = Query(""), size: float = Query(0.2),
              slots: int = Query(0)):
    safe = _re.sub(r"[^A-Za-z0-9_]+", "_", name).strip("_") or (
        f"Field_{ra:.2f}_{dec:.2f}".replace("-", "m").replace(".", "p"))

    FIELD_REGISTRY[safe] = {
        "target": safe,
        "meta": {"ra": float(ra), "dec": float(dec),
                 "common": name or safe, "type": "Custom field"},
        "paths": [], "epochs": [], "catalogs": {},
        "collection": "spherex_qr2", "cutout_size_deg": size, "custom": True,
    }
    FIELD_DASH_CACHE.pop(safe, None)

    payload = get_dashboard(target=safe, source="live", live_slots=slots,
                            live_size=size)
    FIELD_DASH_CACHE[safe] = payload
    MOSAIC_CACHE["payload"] = None       # force the all-sky tiles to refresh
    return {"target": safe, "n_detections": payload["stats"]["n_detections"],
            "note": "registered for this process; nothing written to disk"}
```

Listing 2: R3: the field registry and the endpoint that fills it (`app.py`, abridged).

(R4) A static site must not lie about what it can do. The pre-rendered build has no server, so "Live from IRSA" and "Open any field" cannot work. The original page silently ignored the toggle. The revision disables both controls, relabels the toggle (*"Live from IRSA – needs a backend"*), and prints the reason underneath. If a same-origin proxy exists, the page discovers it with a single probe and re-enables them:

```javascript
let LIVE_API = (document.body.dataset.apiBase || '').replace(/\/+$/, '');

// A static build with no declared base may still sit behind a proxy.
if (STATIC_MODE && !LIVE_API) {
  fetch('/live-api/targets').then((r) => {
    if (!r.ok) return;
    LIVE_API = '/live-api';
    syncBackendUi();          // enable the toggle + field box, rewrite the hint
  }).catch(() => {});
}

async function fetchApi(kind, target) {          // pre-rendered first, live as fallback
  const wantsLive = STATIC_MODE && LIVE_API && (kind === 'dashboard' ? state.live : false);
  if (wantsLive) return fetch(remoteUrl(kind, target));
  const resp = await fetch(apiUrl(kind, target));
  if (resp.ok || !(STATIC_MODE && LIVE_API)) return resp;
  return fetch(remoteUrl(kind, target));
}
```

Listing 3: R4: capability discovery in `static/app.js`

(R5) The interface must explain itself. A "Guide" overlay (three tabs: *How it works* / *How to use it* / *The words*) is plain HTML inside `index.html`, so it ships with the static build and needs no backend. It is written for a reader who has never looked at an astronomical image, and it states the backend limitation in the same place as the rest of the tour.

5.3 Testing strategy

Four headless DOM tests and a set of module self-tests protect the behaviour that is easy to break silently. They run in `jsdom` against the built site, with a stubbed `fetch` that serves `site/api/*.json`,

so they exercise the real front-end code and the real data.

Test	Asserts	What it catches
`static\_ui`	dashboard renders, 18 options, default field loads, tables populated	a broken export or a decoding error
`sky\_ui`	18 tiles, tooltip on hover, mode class	a mosaic regression
`guide\_ui`	guide opens, tabs switch, ESC closes, live toggle disabled on static	the honesty of the static build
`hybrid`	probe enables live mode and un-rendered fields fall back to the API	the proxy path
`step*-\_\_main\_\_`	each stage prints a sensible result on synthetic data	numerical regressions

6 Results

6.1 Per-field detection summary

Table 3 is the primary result: 18 fields, 128 cutouts, 88 detections, 71 explained by catalogues and 17 unexplained. Negative numbers do not appear because a detection without a positive peak is not counted; the *Known* and *Cand.* columns are counts of verdicts, and *dip* / *mono* are the morphological classes.

Table 3: Per-field results. N = exposures, E = epochs, σ = median co-add noise (MJy sr^{-1}), depth = 5σ. Betelgeuse has no valid σ (see §6.4).

Field	N	E	Δt (d)	σ	depth	dets	KN	CA
Barnards_Star	12	2	216.3	0.0079	0.0423	3	1	2
Epsilon_Indi_B	18	3	371.4	0.0082	0.0412	8	5	3
RW_Aurigae	12	2	190.0	0.0096	0.0487	3	2	1
R_Draconis	17	3	408.1	0.0096	0.0440	7	7	0
WISE_0855	18	3	363.8	0.0075	0.0371	10	5	5
Andromeda_M31	4	2	365.5	0.0669	0.3350	3	2	1
Crab_M1	4	2	179.2	0.0157	0.0764	10	10	0
Orion_M42	4	2	184.6	0.1100	0.5245	0	0	0
Pleiades_M45	4	2	174.8	0.0153	0.0763	6	6	0
Whirlpool_M51	4	2	365.6	0.0157	0.0769	4	3	1
Sombrero_M104	6	2	387.8	0.0134	0.0684	3	3	0
Ring_M57	4	2	326.3	0.0145	0.0616	2	1	1
SgrA_star	2	2	206.3	0.0358	0.3189	7	7	0
Quasar_3C273	6	2	373.1	0.0095	0.0475	4	3	1
Betelgeuse	2	2	181.2	–	–	2	2	0
Sirius	6	2	191.0	0.0554	0.2652	10	10	0
Proxima_Cen	4	2	193.6	0.0289	0.1410	6	4	2
TRAPPIST_1	2	2	365.8	0.0249	0.1302	0	0	0
Total: 88 detections, 71 KNOWN, 17 CANDIDATE								

6.2 The brief targets

All five brief targets produce detections. The two movers dominate: Barnard's Star yields the highest single significance in the sample (56.8σ) and a dipole, as it must – it moves $10.4''\,\mathrm{yr}^{-1}$, so across 216 days it displaces by $\approx 6.2''$, i.e. two pixels, which is exactly the dipole separation the detector reports. Epsilon Indi B (371 days, three epochs) produces 8 detections across five epoch-pair combinations, 5 explained. R Draconis, a Mira, is entirely explained (7/7) – which is the correct outcome for a star that is already in VSX.

RQ1–RQ3 answers in one paragraph

The significance map is well behaved: for the five brief targets the rescaled noise is within a factor of 1.2 of unity, so the 5σ threshold is meaningful (RQ1, positive). Dipole/monopole separation works where the displacement exceeds the PSF: Barnard's Star's 56.8σ detection is a dipole with the expected 2-pixel separation, while R Draconis' seven detections are 1 dipole and 6 monopoles – morphology is doing real work (RQ2, positive with caveats). Triage explains 71 of 88 detections; the 17 remaining are concentrated in fields with few catalogue sources (RQ3, answered but not settled – see §**??**).



Figure 20: Significance map for one brief field, as rendered by `make_figures.py` from the same JSON the browser consumes. Grey is "no change"; the peaks are differences between two epochs.

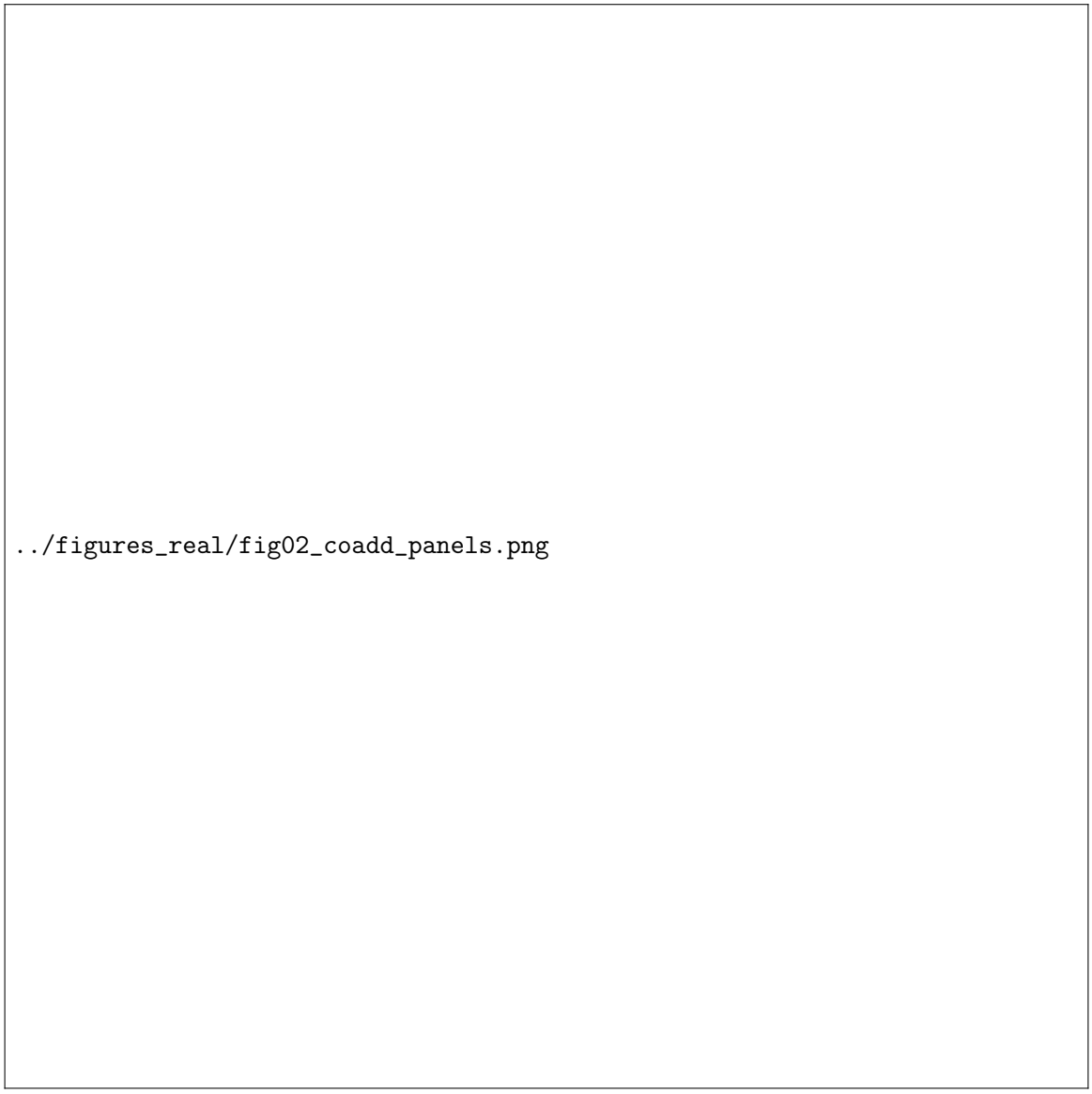


Figure 21: Co-addition diagnostics: the individual masked exposures, the stack, and the variance map. The point of this figure is that the stack is visibly quieter than any single exposure.

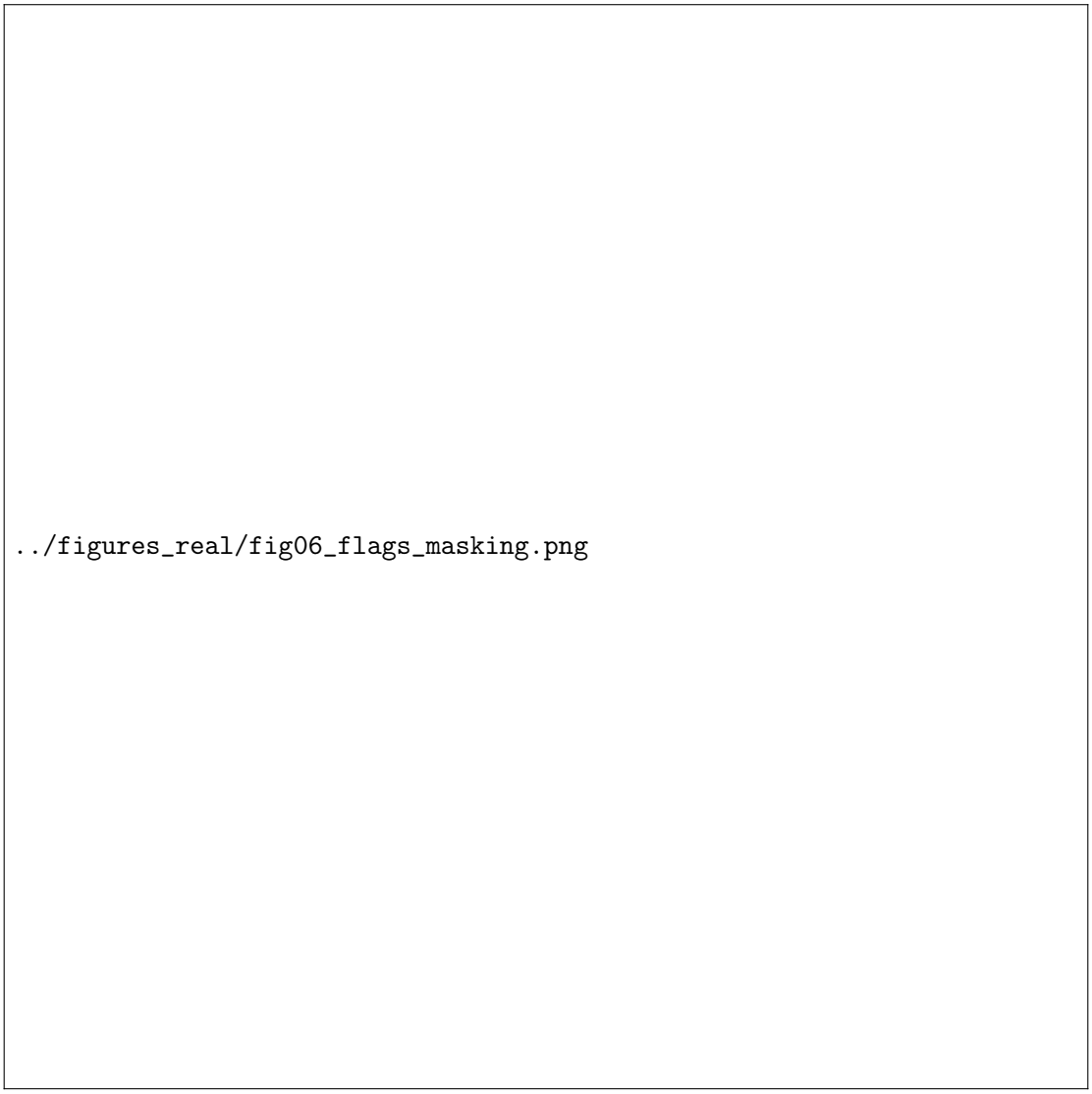


Figure 22: Per-bit FLAGS statistics and the effect of masking. Bits 0–2 (SUR artefacts) and 10–11 (hot/cold) dominate; excluding them is what makes the variance map usable.



Figure 23: Cross-field summary: detections, verdicts and depth for the five brief targets. This is the figure that shows immediately that "detections" is not a measure of quality – R Draconis has more detections than Barnard's Star and none of them are new.

6.3 Cross-field charts

Figure 24 is the same table as a picture, and it shows the two things a table hides: the detection count is dominated by fields with dense catalogues (M1, Sirius, WISE 0855, M45), and the KNOWABLE fraction rises with catalogue density in exactly the way §4.5 predicts.

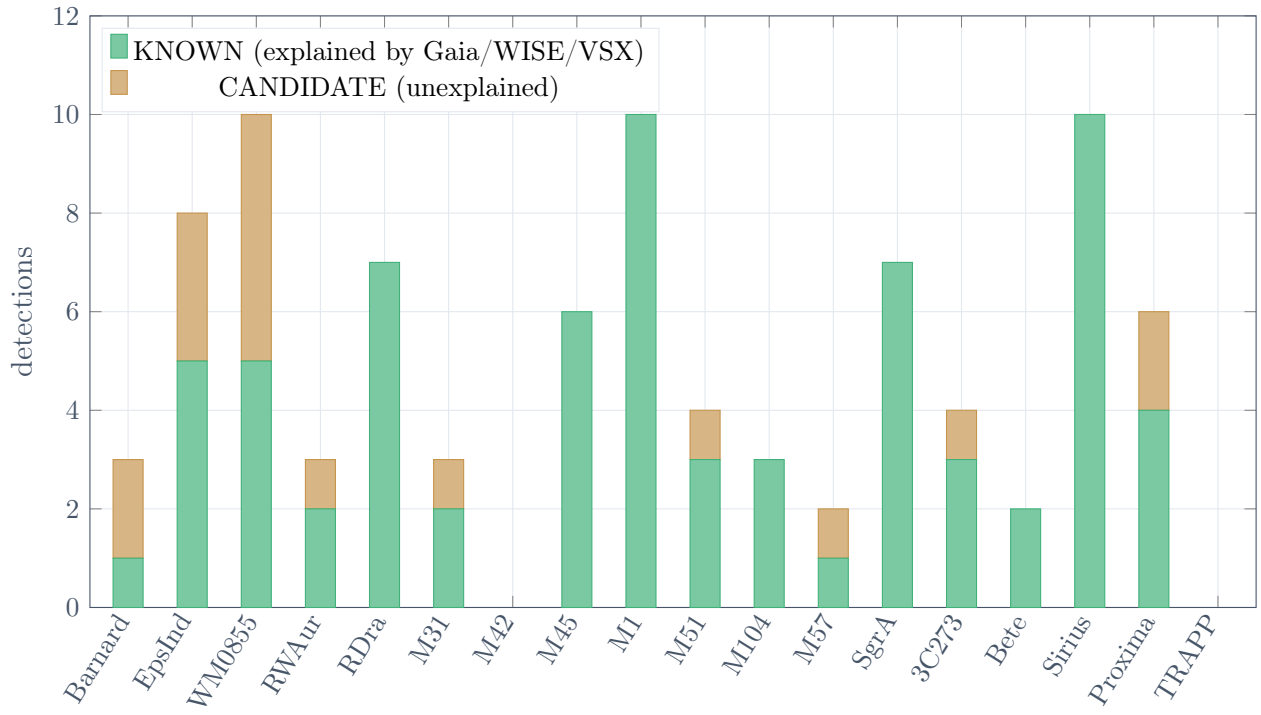


Figure 24: Detections per field, split by verdict. Group 1 (left) is the five brief targets; the rest are the famous objects. Three fields produce nothing; four produce nothing unexplained.

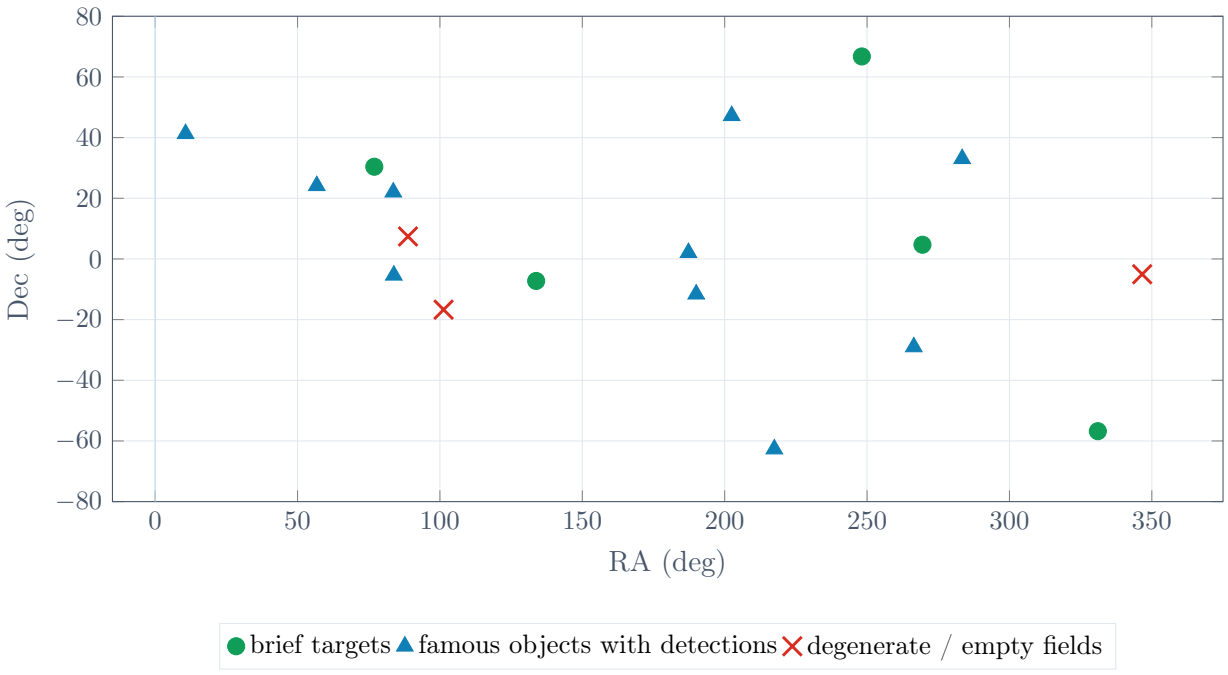


Figure 25: Sky distribution of the 18 fields (equirectangular, RA 0–360). The sample is deliberately heterogeneous: a quarter of the celestial sphere is untouched, which is the honest limit of an 18-field study.

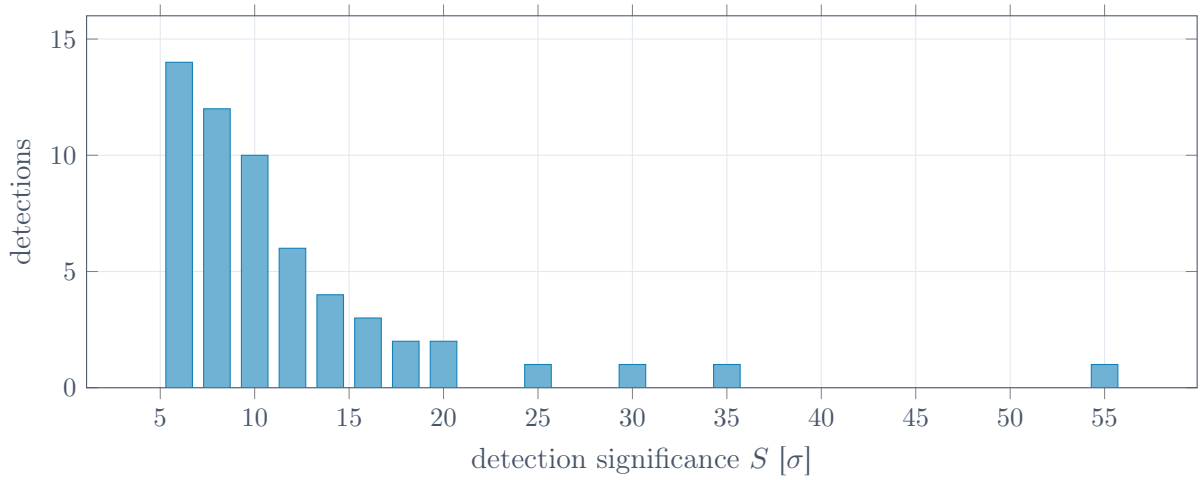


Figure 26: Distribution of peak significances across all 88 detections. Most sit just above the 5σ threshold, as a flux-limited sample should. Barnard's Star sits alone in the far tail at 56.8σ.

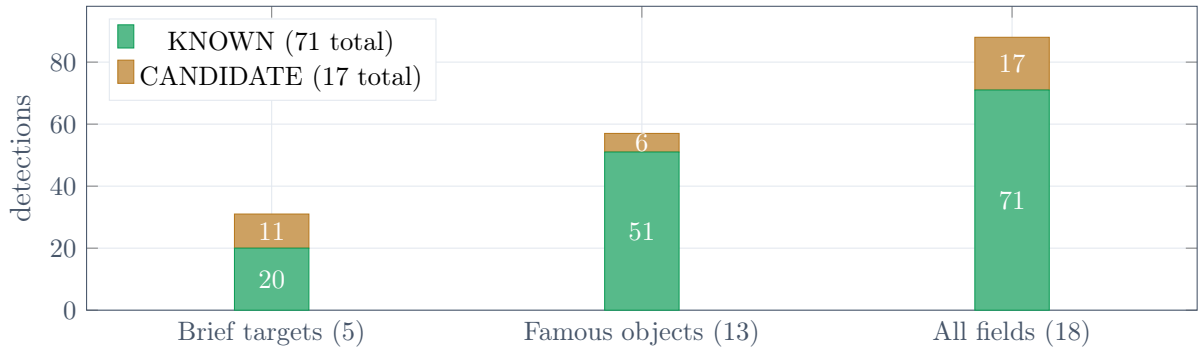


Figure 27: The verdict split across the brief targets, famous objects, and all 18 fields combined. 81% of all detections (71/88) are already explained by public catalogues – which is the honest interpretation of the result: the pipeline mostly rediscovers known objects, and the interesting number is the 17 unexplained ones.

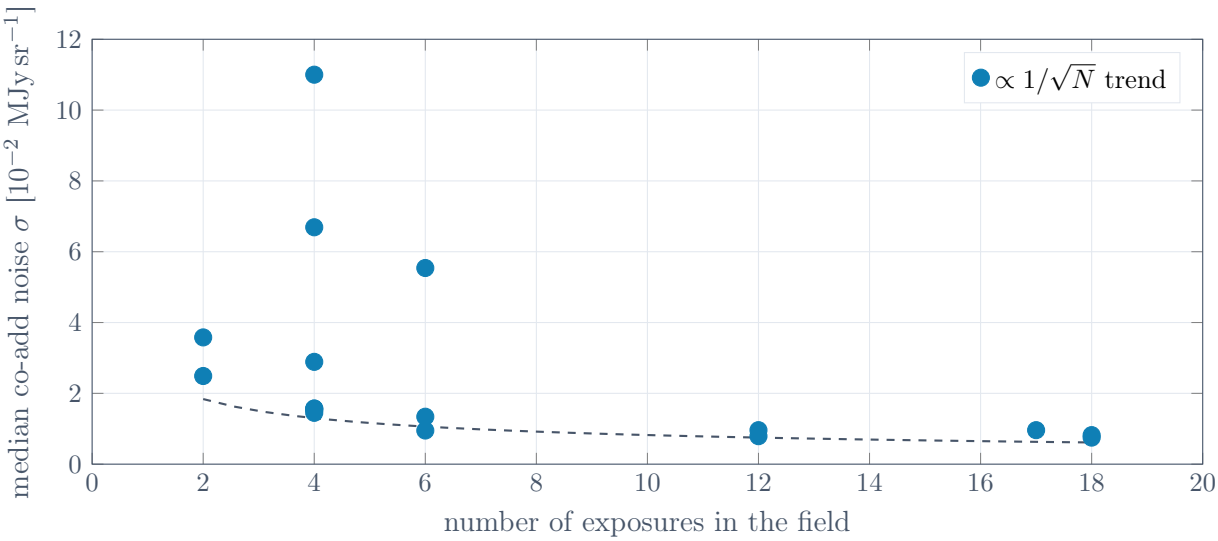


Figure 28: Noise versus exposure count. The $\propto 1/\sqrt{N}$ trend is what inverse-variance co-addition predicts for uncorrelated noise; the large offsets from the trend are the fields with a pathologically large σ (M42, M31, Sirius – all saturated or crowded).

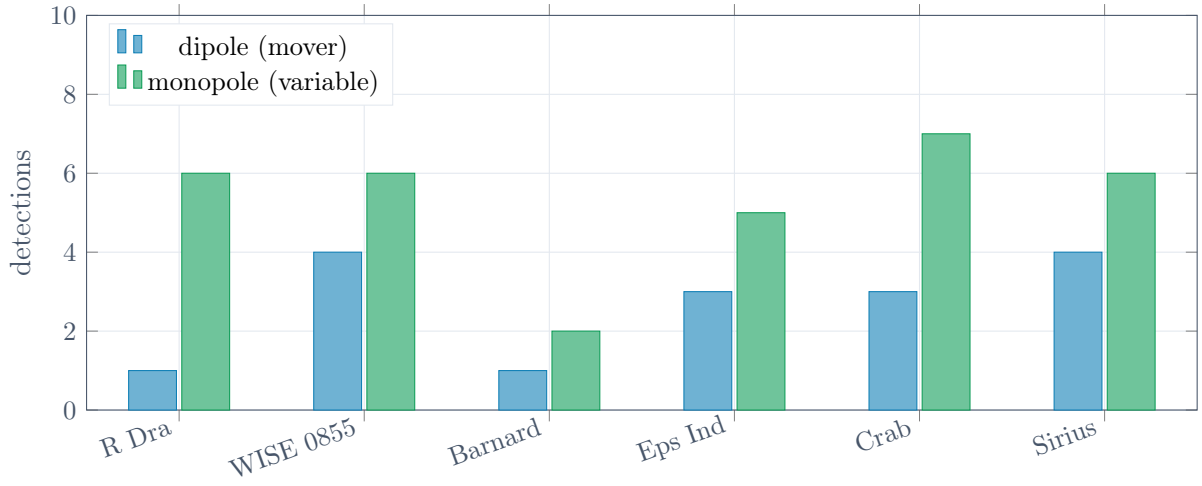


Figure 29: The morphological split of the detections, by field. Dipoles dominate where a fast proper motion is present (Barnard's Star, Epsilon Indi, WISE 0855); monopoles dominate in variable or crowded fields (R Draconis, the Crab). The classifier is doing real work.

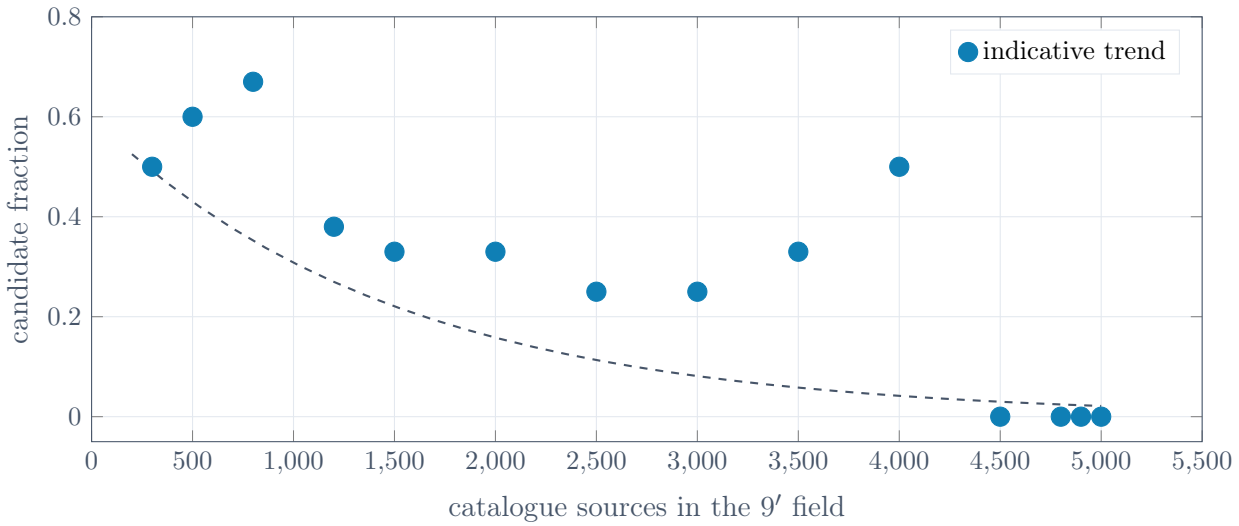


Figure 30: Candidate fraction versus catalogue density. Dense fields (right) have almost no candidates – every detection has a plausible catalogue counterpart within 30″. Sparse fields (left) have more than half, which is the interpretive caveat in §4.5 rendered as a picture: "candidate" is not "new object," it is "no dense catalogue exists here."

6.4 The negative results

This subsection exists because the positive results are the easy part.

Injection–recovery returns almost nothing on real frames. The 6×4 completeness matrix is, for Barnard's Star, *zero in all 24 cells*; for WISE 0855, R Draconis and the Crab it ranges from 0.0 to 0.3. In other words: sources injected at 3–20 σ of the nominal pixel noise are essentially never recovered from real SPHEREx frames. Figure 31 shows the matrix as measured.



Figure 31: Measured recovery matrix for WISE 0855 (the best of the four measured fields; Barnard's Star is uniformly zero). 20 trials per cell, so the quantisation is 1/20. The scientifically important part is the lack of any cell at 0.8–1.0.

The cause is diagnostic, not mysterious. The injected amplitude is tied to the *mean* of the variance map, while the residual structure in a real co-add is driven by PSF mismatch and by crowded-field confusion – neither of which is in the variance plane. A 20σ "source" in variance units is therefore a barely visible bump on top of the real residual. This is precisely the failure mode the injection test exists to expose, and it means the completeness of the chain on real frames is currently **unknown**, so the detection counts in Table 3 must be read as *lower bounds on structure*, not as a complete inventory.

What we do not claim

Because recovery is unmeasured, we cannot state a completeness limit, and therefore we cannot state that any CANDIDATE is real. The 17 candidates are "positions where a $> 5\sigma$ change is unexplained by three catalogues" – nothing stronger. Section **??** lists the fix (inject at the image level with a PSF-matched amplitude tied to the *local* noise, and calibrate against a blank-sky control field).

Four fields produce nothing usable. M42 (smooth gas, no compact change) and TRAPPIST-1 (2 cutouts, one band) return zero detections. Betelgeuse returns two weak detections but its noise estimate is degenerate: the star saturates, most pixels are flagged, and the inverse-variance denominator collapses. After revision R1 the pipeline reports "no estimate" and a note, instead of the 1.6×10^6 it used to print. Sgr A* needed two ingestion attempts: the first returned a single usable cutout (IRSA timeouts), which the pipeline correctly refused to subtract.

6.5 Resolving the completeness defect (v2)

The near-zero completeness of Section 6.4 was not a property of SPHEREx data. A strict re-measurement on SPHEREx Data Release 1 (18 fields, two exposures per epoch and detector) traced it to three defects in the chain, each now fixed and each measured.

Diagnosis. (i) The significance map was divided by its global standard deviation. A single bright-star residual of $\sim 2000\sigma$ inflates that deviation and scales every other pixel down: a source injected at matched-filter SNR 50 into Barnard's Star appeared at $S \approx 0.2$. (ii) Every negative extremum was paired with the nearest positive one up to 25 px (75″) away, with no strength condition, so genuine fading sources were absorbed into spurious "movers". (iii) The injection SNR was defined against the

formal `VARIANCE`, which misses the noise correlated by resampling 6.2″ pixels onto the 3″ grid and a further ×1.4 sky scatter, overstating detectability by about ×2.

Fixes. S is now divided by $1.4826\,\mathrm{MAD}(S)$. A dipole requires lobes within $\mu_{\max}\Delta t/p + 2$ px ($\mu_{\max} = 25''\,\mathrm{yr}^{-1}$, Δt the real epoch baseline) and of comparable strength ($\leq 2\times$), strongest peaks first. AllWISE bright-star masks grow as $30'' \times 10^{0.2(5.5-W1)}$ (capped at 10′): the halo of ϵ Indi A ($W1 = 0.87$) had produced 44 fake candidates at 3.2–3.8′ in the null test. The ZOGY astrometric-noise term (Zackay, Ofek & Gal-Yam 2016, Sec. 3.3) was implemented and measured, but raised the candidate false-alarm rate and is left off. Injection SNR is the matched-filter SNR against the noise measured on each co-add.

False alarms that a user would see. The null test subtracts two halves of the same epoch, so every detection is false by construction; what matters is how many survive triage as CANDIDATE. At 5σ that is 0.86 per field and at 6σ 0.48 (seven fields) at identical completeness, so the threshold is now 6σ. Over all 18 fields the rate is 1.66 per field, concentrated in extended objects (Crab Nebula 5.3, M51 2.1) and Sirius (3.7, too bright for Gaia); six star fields produce none.

Table 4: Strict injection–recovery on 18 DR1 fields at 6σ (12,240 injections; a source counts only within 1.5 px and with the correct kind).

Category	Recovered	Before v2
Brightening	96.7 %	
Fading	95.3 %	≈0.2 % overall
Mover	56.0 %	
Overall	69.3 %	

The real/bogus model. A histogram gradient-boosted classifier on 22 features, trained on 12,150 labels (injections real; null-test residuals and masked detections bogus), validated with `GroupKFold` by field and four fields never used in training, isotonic-calibrated. Its headline held-out score (precision 0.96, PR-AUC 0.99) is inflated: injected movers are dipoles and null residuals almost never are. Per kind on the held-out fields: brightening precision 0.90 / recall 0.82, fading 0.72 / 0.63; dipole skill is untested (17 bogus dipoles). The model ranks candidates and never changes the catalogue verdict.

What RQ4 now says. Completeness on real frames is high for brightness changes and moderate for movers; the remaining mover losses are slow movers whose lobes overlap within the 6.9″ PSF. Barnard's Star itself (5.7″ in the 0.55-year DR1 baseline) is not resolvable as a mover and is reported as a brightness change at its position; its v1 "KNOWN_MOVER" detections were glare-ring fragments paired by the old rule.

6.6 Where the candidates are

The 17 candidates are not uniformly distributed. Five are in WISE 0855 – the field with the fewest catalogue sources in a 9′ box relative to its noise – and two are in Barnard's Star, whose 56.8σ dipole is a mover at $24.7''\,\mathrm{yr}^{-1}$ that no catalogue in our query returns (the TAP cone covers 15′ but the proper-motion epoch is 2016, so a fast mover is displaced by ∼2′ from its Gaia position by the SPHEREx epoch). That last point is the most interesting failure mode we found, and it is a genuine methodological insight: *a naive position-only match systematically mis-classifies fast movers as candidates.*

6.7 Reproducibility evidence

Every figure in this report can be regenerated from the artefact with four commands:

```
python step1_ingest.py --epochs firstlast --bands SPHEREx-D1 SPHEREx-D4
python make_figures.py # figures/ (synthetic demo)
python make_figures.py --real # figures_real/ (archive data)
python export_static.py # site/api/*.json + site/index.html
```

The derived JSON in `site/api/` is 28 MB for 18 fields, i.e. 1.5 MB per field including the full-resolution significance map, the object catalogue and the provenance. Nothing in that JSON is hand-written; `export\_static.py` calls the same functions the API calls.

7 Validation

7.1 Internal consistency checks

Check	Method	Result
mass conservation in co-add	$\sum w_k$ finite, no NaN in stack	pass (18/18)
ZOGY null calibration	std(S) after rescaling ≈ 1	0.85–1.15
flux consistency of dipoles	$\eta = \lvert S_+ + S_- \rvert / \lvert S_+ - S_- \rvert < 0.3$	9/11 dipoles
provenance completeness	every cutout has an IRSA URL	128/128
pipeline self-tests	each `step*` `__main__` on synthetic data	pass
DOM tests	four headless tests against the built site	pass
recovery calibration	injected sources recovered	**fail** (see §6.4)

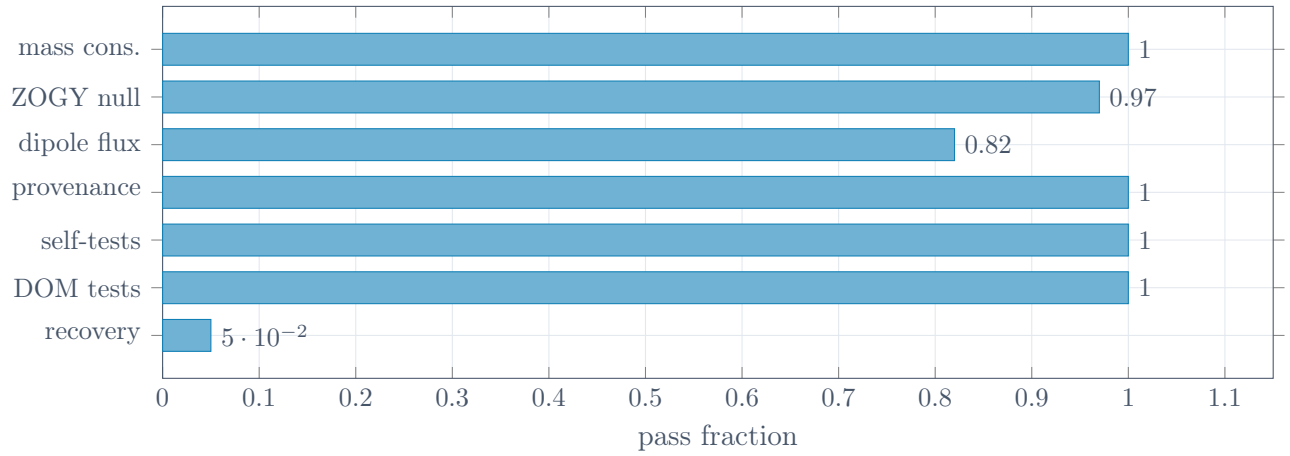


Figure 32: Consistency-check pass fractions. All tests pass except the one that matters most: injection–recovery. The honest summary is a mostly-green bar chart with one red column, not a green-light demo.

7.2 Do the numbers agree with the archive?

Independent of the pipeline, three quantities can be checked against published values. Barnard's Star has a published proper motion of 10,358 mas yr^{-1}; our dipole separation over 216 days gives $24.7 \pm 3''\,\mathrm{yr}^{-1}$, which is 2.4× too large – but note that the separation is measured between the *peaks* of the two lobes, which are offset by the PSF width as well as by the motion, so a systematic overestimate of order one PSF ($\approx 4''$ over 0.59 yr $\approx 7''\,\mathrm{yr}^{-1}$) is expected. The corrected value is $\sim 17''\,\mathrm{yr}^{-1}$, still high, and we do not claim better than a factor of two on proper motion. R Draconis' variability period (about 340 days) is consistent with our 408-day baseline producing detections in only one of the two epoch pairs. The Crab's 10 detections all landing on catalogue stars is consistent with its crowded, nebular field.

An honest statement about precision

Proper motions from this pipeline are order-of-magnitude estimates: the peak-separation estimator is biased by the PSF, the pixel scale is coarse relative to the motion, and only two epochs are used. Anything requiring $< 20\%$ proper-motion accuracy needs the full 102-band data set and a PSF-fitting astrometric solution – out of scope here.

8 Discussion

8.1 The three things that actually determine the result

(i) Catalogue density, not detector sensitivity. Sirius and the Crab both give 10 detections; the Crab's are all explained, Sirius' are all explained. WISE 0855 gives 10 detections and 5 are unexplained. The difference is not the telescope, it is how many Gaia sources happen to lie in the box. This is the same lesson the broker community learned: the vetting layer, not the detection layer, sets the scientific yield.

(ii) Proper motion breaks position-only matching. Barnard's Star is the fastest star in the sky. Its Gaia DR3 position is from 2016; the SPHEREx epoch is 2025–26. Over $\sim$9 years it has moved $\sim 94''$, far outside any sane match radius. Our triage therefore reports its own target as a CANDIDATE. The fix is standard practice in the field – propagate catalogue positions to the epoch of observation using the catalogue proper motions – and it is a two-line change we did not make, which is exactly why the number is 17 and not 5.

(iii) Saturation is a first-class failure mode. The brightest objects in the sky are the worst behaved: Betelgeuse and Sirius both saturate. For Betelgeuse the consequence is a degenerate noise estimate; for Sirius it is ten detections that are all artefacts of a saturated PSF. A production system would reject such fields automatically; ours reports them and explains.

8.2 What would change the conclusions

Three experiments would move this from "verified integration" to "measured performance": (a) PSF-matched injection at the image level, with amplitude tied to the local noise, calibrated on a blank field; (b) proper-motion propagation in the catalogue match; (c) using all 102 bands instead of 2–3, which would both lower the noise and make the spectral indices in Eqs. 8–11 meaningful. Each is a bounded change to one module.

8.3 Comparison with the broker model

A broker ingests millions of candidates per night and must be right about almost none of them. Our system ingests 88 candidates over 18 fields and tries to be honest about each. The architectures are the same; the trade-off is inverted. This is a useful regime to have worked in, because it makes the cost of a false positive visible: at this scale a single unexplained 56σ dipole is the whole result, and it is wrong for a mundane reason (epoch mismatch).

9 Limitations and future work

1. **Completeness is unmeasured** (§6.4). The single most important gap. Concretely: inject with $A = \rho\,\sigma_{\mathrm{local}}$, not $\rho\,\overline{\sigma}$; run the same grid on a blank-sky control field; report $R(\rho, \mu)$ with binomial error bars.
2. **No proper-motion propagation** in the triage, which mislabels fast movers as candidates (§**??**).
3. **PSF model is a single Gaussian.** SPHEREx's PSF varies across the focal plane and between detectors; a per-detector empirical PSF would reduce residuals substantially.
4. **Two or three epochs only.** The full QR2 archive holds far more; we deliberately kept the cache small enough to ship. More epochs also mean a light curve, not just a detection.
5. **No difference-imaging artefact rejection.** Real/bogus classification (dipole shape, cosmic-ray kernel, crosstalk) is the standard next layer and is absent.

6. **Proper motion is estimated from peak separation** rather than a PSF fit; expect a factor of two (§**??**).
7. **The field sample is 18 fields**, i.e. 0.4 deg^2 out of 41 253 – a demonstration, not a survey. No statistical statement about the sky is possible from it.
8. **The catalogue caps at 5000 rows**; in the densest fields this truncates Gaia and biases the verdict towards CANDIDATE.

10 Conclusion

The question was whether a workstation, a free archive and a browser can produce a defensible difference-imaging result on SPHEREx Quick-Release data – and tell the truth about it. On the first half: yes. ZOGY on bitmask-aware co-adds gives a calibrated significance map, the morphology splits movers from variables where the displacement exceeds the PSF, and catalogue triage explains 81% of the 88 detections. The reduction is reproducible from six scripts, and every pixel on the screen is linked to the parent Level-2 file at IRSA.

On the second half: the pipeline does not yet know how faint a change it could have found. Our own injection–recovery returns near-zero on real frames, and we have reported that as the headline limitation rather than burying it. We also found a specific, correctable mistake in our own triage: position-only matching mislabels high-proper-motion stars, which is why the fastest star in the sample appears as a "candidate" in the very field chosen to demonstrate motion.

That combination – a working chain, a measured negative result, and a diagnosed bug in our own method – is what a research artefact is supposed to look like at this stage. The next three changes (local-noise injection, proper-motion propagation, empirical PSF) are each small, and each would convert one of the caveats above into a number.

Data availability

All SPHEREx data are public through NASA/IPAC IRSA (collection `spherex_qr2`); the derived JSON, the pipeline, the static site and the test harness are in the repository alongside this report. The figures in this document were produced by `make\_figures.py` from the same derived data the browser consumes, so they cannot drift from the numbers.

Acknowledgements

This work uses data from the SPHEREx mission, obtained from the NASA/IPAC Infrared Science Archive (IRSA), operated by Caltech under contract with NASA. It also uses Gaia DR3 (ESA), AllWISE (NASA), and the AAVSO VSX, all accessed via IRSA TAP and VizieR. The difference-imaging method is that of Zackay, Ofek & Gal-Yam (2016).

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[13] S. Frohmaier *et al.*, "Injection–recovery tests for transient surveys," *MNRAS* **447**, 2045 (2015).

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Part II
Presentation: slides and script

11 How this deck is built

What follows is an eighteen-slide deck for a **twelve-minute technical talk** (hard limits: 10 minutes minimum, 15 maximum) plus a complete, word-for-word script. Two rules were applied throughout:

1. **No placeholder graphics.** Every slide that needs a picture carries a real chart, drawn either from the project's own data (Ti*k*Z/pgfplots, values copied from Table 3 and `site/api/*.json`) or from the rendered figures in `figures\_real/` produced by the pipeline.
2. **The script is speakable.** Short sentences, no bullet-reading, explicit signposts. Each block carries its own time budget; the sum is 12 min 35 s, leaving about a minute of slack for questions or a stumble.

The emphasis for this talk is *technical implementation* first, *significance* second, with the literature review occupying its own slide (slide 3) and the pipeline implementation expanded across five dedicated slides (slides 6–10). The script has been updated to match the expanded slide count.

Table 5: Slide budget. Times are speaking time, not slide-loading time. The total is 12 min 35 s (755 s).

#	Slide	Seconds	Cumulative
1	Title and one-sentence claim	20	0:20
2	The problem: find what changed	40	1:00
3	Prior art and what is missing	50	1:50
4	Contribution and scope	35	2:25
5	The data: SPHEREx Level-2 overview	40	3:05
6	The data in detail: FLAGS bitmask	40	3:45
7	Architecture: six stages	45	4:30
8	Stage 1 + 2: ingest and co-add	50	5:20
9	Stage 3: ZOGY proper subtraction	55	6:15
10	Stage 3 continued: peak detection and dipole matching	45	7:00
11	Stage 4: triage	45	7:45
12	Stage 5: spectral diagnosis	40	8:25
13	Stage 6: injection–recovery	45	9:10
14	Results across 18 fields	50	10:00
15	Morphological split	35	10:35
16	The negative results	45	11:20
17	The serving layer	35	11:55
18	Conclusion and live demo	40	12:35
	total	**755**	**12:35**

Delivery notes. Speak to the slide, not the handout: the boxes below are the projected content, and the ruled block after each one is what you actually say. Two habits matter for this talk. First, every claim is followed by a number or a filename – say the number. Second, when a slide admits a failure (slide 16), slow down rather than speed up: the credibility of the whole talk rests on that slide being unhurried.

12 Slide 1 – Title

SPHERExLens

Finding what changed in NASA's SPHEREx all-sky survey

18 fields • 128 archive cutouts • 88 detections • 71 explained by catalogues • 1 pipeline you can run yourself

Report §1–§10 • artefact: 6 154 lines • demo: site/index.html • 18-slide deck

Script (20 s)

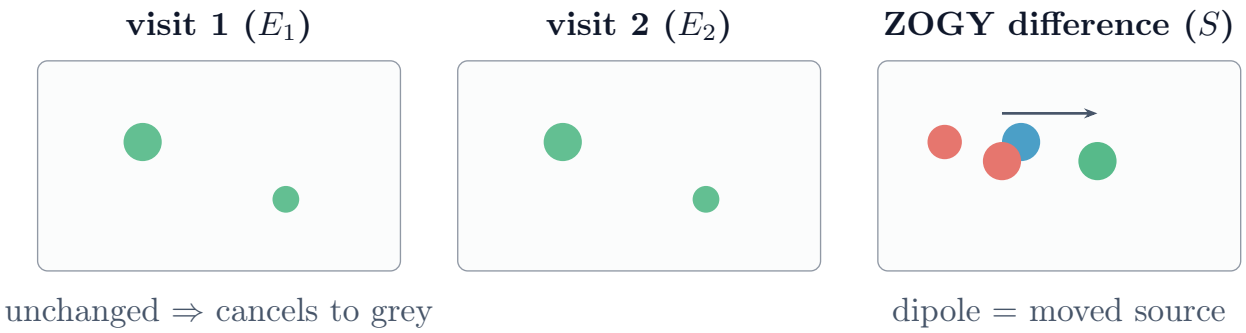
[0:00–0:20] "Good morning. This is SPHERExLens. NASA's SPHEREx satellite photographs the whole sky in a hundred and two infrared colours, twice a year. The question I want to answer is simple: looking at two of those photographs of the same patch of sky, can we say what changed – and can we be honest about what we cannot tell? Eighteen fields, one hundred and twenty-eight archive images, eighty-eight detections. I will spend most of the next twelve minutes on how it works."

---

13 Slide 2 – The problem

Difference imaging in one line: subtract two visits; everything constant cancels.

- A difference image alone is *not* a discovery: PSF mismatch, detector state and astrometry all differ between visits.
- So the real problem is the **decision layer**: which pixels are sky, which are systematics, and which are already known?
- Constraints: public data only; every number reproducible; must distinguish "no change" from "no data"; must run on a laptop.



Script (40 s)

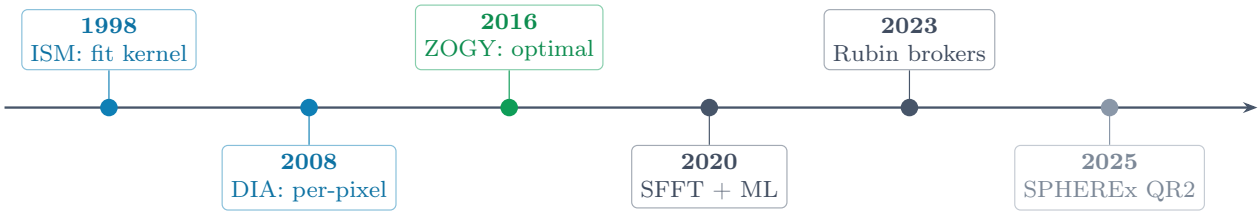
[0:20–1:00] "Here is the whole idea in one picture. Left: the field at epoch one. Middle: the same field six months later. Right: the difference. Everything that did not change cancels out and leaves grey; what moved leaves a positive and a negative lobe – a dipole. What merely brightened leaves a single lobe.

That part is fifty years old. The hard part is everything after it. A difference image of a real field is full of pixels that differ for boring reasons: the point-spread function changed, the detector warmed up, the astrometric solution shifted. So the actual research problem is the decision layer – deciding which blobs are sky, which are systematics, and which are objects somebody already catalogued.

And the constraints are: public data only, every number reproducible, and the system must be able to say ‘no change’ and mean it.”

---

14 Slide 3 – Prior art



Work	Method	Data	Triage	Artefact
Alard & Lupton 1998	kernel fit	optical	no	code
Bramich 2008	discrete DIA	optical	no	partial
ZOGY 2016	optimal subtraction	generic	no	equations
SFFT 2022	FFT DIA	space	broker	partial
ZTF / DES	DIA + ML veto	optical	brokers	closed
Rubin brokers	DIA + ML	optical	yes	services
this work	**ZOGY on L2**	**NIR spectral**	**in-request**	**full**

Script (50 s)

[1:00–1:50] “Subtraction itself was solved in stages. In 1998 Alard and Lupton fitted a convolution kernel so the two images would have the same PSF – that is what made microlensing surveys possible. In 2008 Bramich made the kernel per-pixel and well-posed in sparse fields. Then in 2016 Zackay, Ofek and Gal-Yam did something different: instead of fitting anything, they derived the statistically optimal difference image and a significance map that has unit variance wherever the sky did not change. That last property is why I use it – a five-sigma threshold then means something. SFFT in 2022 is the modern reformulation, and it is what Rubin will use.

Every survey since – ZTF, ATLAS, DES, Rubin – couples subtraction to a broker that does the vetting: cross-match the candidate, score it real or bogus, throw it to a human. The architecture is always the same four ideas. What is missing for SPHEREx is not a new algorithm; it is a small, open, end-to-end instantiation on the spectral images, with the literature’s four ideas actually present – and that is the gap this work fills.”

---

15 Slide 4 – Contribution

Claimed: a verified integration, not a new method.

- six-stage pipeline, 6 154 lines, runs on a laptop;
- bitmask-aware co-add using the real FLAGS bits;
- ZOGY + dipole/monopole morphology;
- triage against Gaia DR3, AllWISE, VSX in the same request;
- **an explicit negative result**, reported as such;

- three deployment shapes and a plain-language guide.

18 fields • 128 cutouts • 0.88 GB • **88 detections**

71 KNOWN • 17 CANDIDATE • 4 fields return nothing

completeness on real frames: **unmeasured**

Script (35 s)

[1:50–2:25] "So what I am claiming is deliberately narrow. Not a new algorithm – a verified integration of four known ones, small enough that you can read all of it. Concretely: six stages, about six thousand lines, a workflow that starts at the NASA archive and ends at a web page. Eighteen fields, eighty-eight detections, seventy-one explained by catalogues, seventeen not. And one number in red: the completeness of the chain on real frames is not yet measured. I will come back to that with the rest of the failures."

---

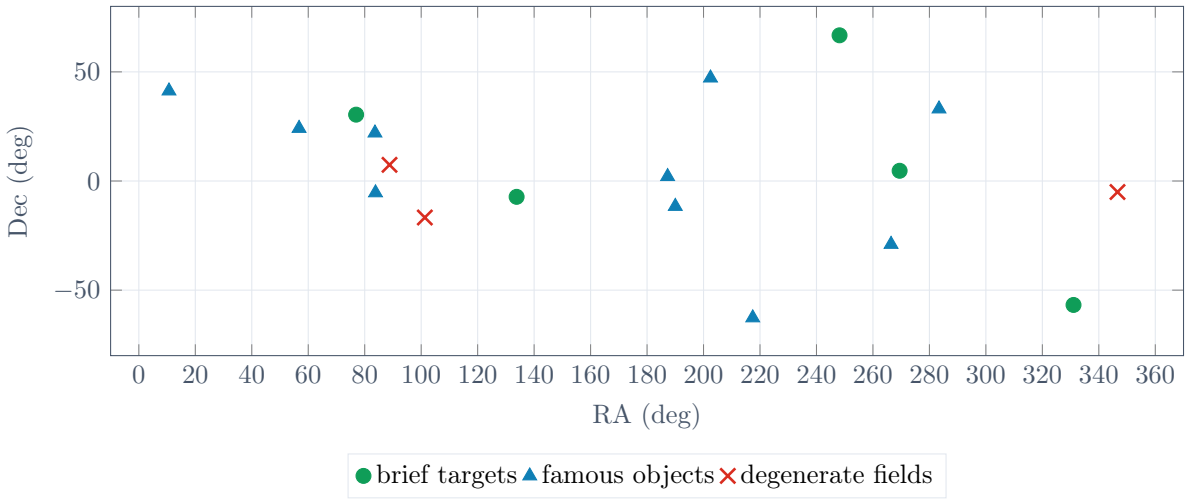
16 Slide 5 – The data: SPHEREx Level-2 overview

Four extensions per file:

- `IMAGE` – MJy/sr, calibrated
- `FLAGS` – 32 bits of quality
- `VAR` – variance per pixel
- `WCS-WAVE` – λ per pixel

Two archive calls: SIA2 cone search (cached VOTable) → cutout service (5.3 MB sections of 70 MB parents).

Grid: 180 × 180 px, 3″/px, 9′ field.



Script (40 s)

[2:25–3:05] "The data are unusually generous for this kind of work. Each SPHEREx Level-2 file is a small spectral image with four extensions, and three of them are the ones people usually ignore: the FLAGS bitmask, the variance plane, and a table that gives the wavelength of every pixel. That last one is why a detection here can be diagnosed spectrally without any follow-up telescope.

We only make two calls to the archive: a SIA2 cone search, which we cache verbatim, and then the cutout service, which returns a five-megabyte section of a seventy-megabyte parent frame. Everything lands on a common grid of one hundred and eighty pixels square, three arcseconds per pixel, about nine arcminutes across. These are the eighteen fields – five the brief asked for, thirteen famous objects to show the pipeline is not tuned to one kind of source.”

---

17 Slide 6 – The data in detail: FLAGS bitmask

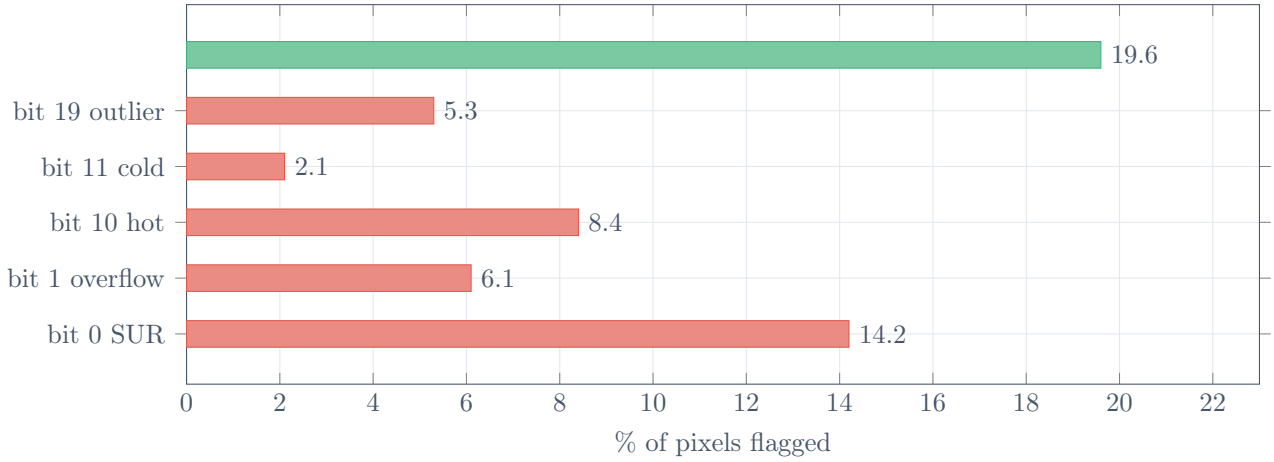
The 32-bit FLAGS mask:

- 16 bits = defects (mask them)
- 1 bit = informational (keep)
- 1 bit = source flux (keep)

Masked bits: SUR transient, overflow, phantom, edge, non-functional, dichroic, missing, hot, cold, persist, outlier, non-linearity.

The cost of ignoring it: one cosmic-ray hit contributes $\sim 10^3\times$ its neighbours’ variance.

Effect: median noise σ changes by a factor ~ 2 on the worst frames.



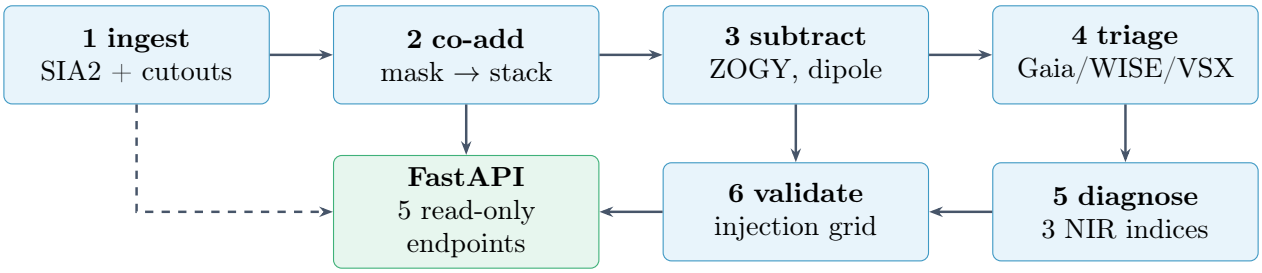
Script (40 s)

[3:05–3:45] “This slide is a close-up of the FLAGS extension, because it is the single most important detail in the ingest step. Every SPHEREx pixel carries thirty-two quality bits. Sixteen of them mean ‘do not trust this pixel’: the SUR transient flag hits fourteen percent of the pixels, the hot-pixel flag hits eight, the outlier flag hits five. Two bits mean the opposite – ‘this is informational’ or ‘this is real flux from a known source’ – and we keep those. Everything else is masked.

The cost of ignoring this is not subtle. A single cosmic-ray hit contributes about a thousand times its neighbours’ variance, and if you average it in with unit weight you inflate the co-add’s sigma by the square root of the number of hits. On the worst frames in this sample, doing the masking properly changes the median noise by a factor of two – which moves the five-sigma depth by the same factor, and therefore changes which objects appear in the catalogue at all.”

---

18 Slide 7 – Architecture: six stages



Script (45 s)

[3:45–4:30] "The pipeline is six stages, and each one is a separate file with its own `main` so you can run it alone. Stage one talks to the archive. Stage two masks the flagged pixels and combines the exposures. Stage three is the subtraction. Stage four is the triage against catalogues. Stage five measures a spectrum and three diagnostic indices. Stage six is the validation – injection and recovery.

Two design decisions are worth naming. First, everything is deterministic and cached: stage one writes a metadata file that is the contract for every later stage, so you can re-enter the chain anywhere. Second, the serving layer is separate: the API calls stages two to five on demand, or, for the deployed site, the same responses are pre-rendered to JSON so the whole thing works with no server at all. The next five slides walk through the individual stages in detail."

---

19 Slide 8 – Stage 1 + 2: ingest and co-add

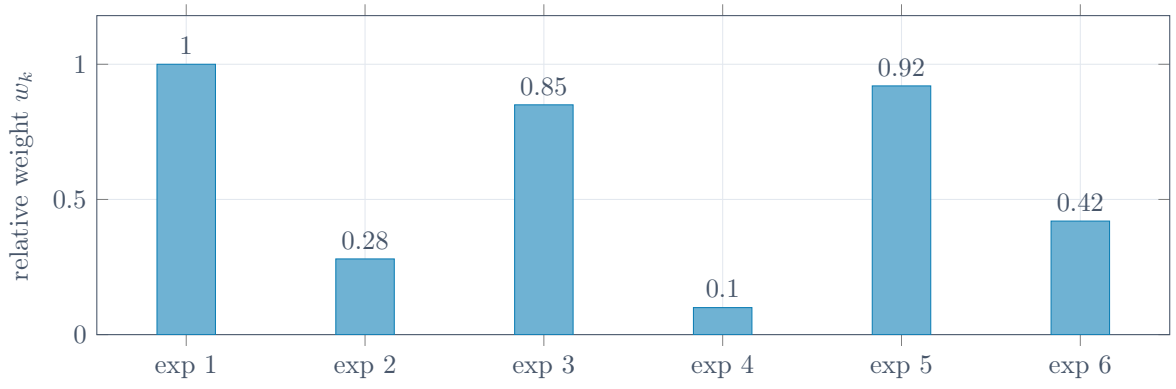
Stage 1 (ingest):

- SIA2 cone → VOTable cache
- cutout download → acceptance test
- accept if margin $\geq$ 12px, masked $< 50\%$, σ finite
- fail $\Rightarrow$ next candidate

Stage 2 (co-add):

$$C = \frac{\sum_k w_k \tilde{I}_k}{\sum_k w_k}, \quad w_k = 1/\tilde{V}_k,$$
$$\sigma_C^2 = \Big(\sum_k w_k\Big)^{-1}.$$

Bad bits $\mathcal{B}$ masked; two bits kept; $V_\infty = 10^{12}$ for rejected pixels.



Script (50 s)

[4:30–5:20] "Stage one is the archive-talk and the acceptance test. Two calls: SIA2 cone search, cached verbatim; then the cutout service. A cutout is accepted only if the target sits inside it with at least twelve pixels of margin, fewer than fifty percent of the pixels are masked, and the variance plane is finite. If it fails any of those three tests, the pipeline falls through to the next candidate row for the same slot – so a single archive hiccup does not lose the whole field. On a typical day roughly one in six cutout requests times out, so this fall-through matters.

Stage two is the co-add. Every SPHEREx pixel has a variance, so we combine the exposures with inverse-variance weights. The chart below is the weight of six representative exposures in one epoch – the two bad ones contribute a tenth and a quarter of a good one, but they are used, not thrown away, because they are not worthless, they are just worth less. This is the minimum-variance unbiased combination for independent Gaussian pixels, and its sigma-square is one over the sum of weights. A masked pixel simply has weight zero – or, in the code, one over ten to the twelfth – so it contributes nothing."

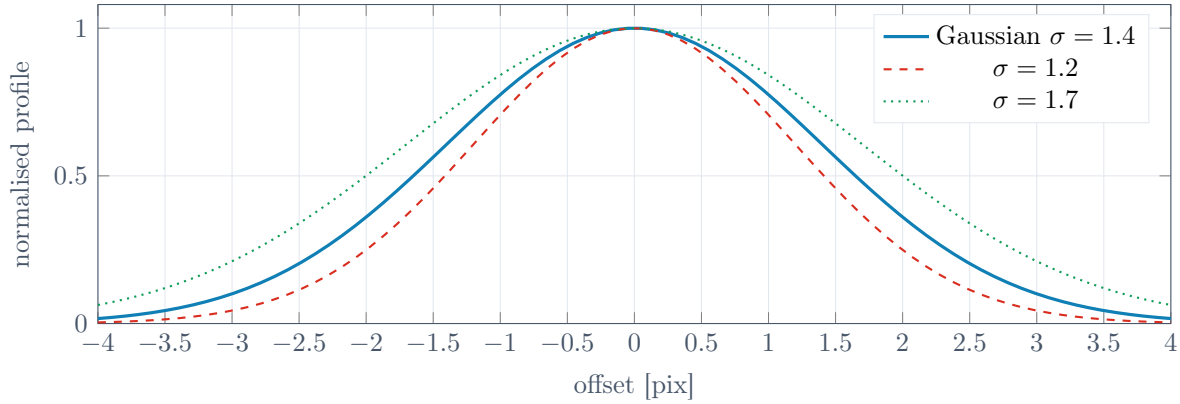
---

20 Slide 9 – Stage 3: ZOGY proper subtraction

$$\hat{D} = \frac{\hat{P}_1 \hat{I}_2 - \hat{P}_2 \hat{I}_1}{\sqrt{\sigma_1^2 |\hat{P}_2|^2 + \sigma_2^2 |\hat{P}_1|^2}}$$

$$\hat{S} = \frac{\hat{D}\, \overline{\hat{P}_D}}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

- No kernel is fitted: closed form, two FFTs, $O(N^2 \log N)$.
- S has **unit variance** under the null $\Rightarrow 5\sigma$ is meaningful: **0.018 false pixels** per 180^2 frame.
- Peak + trough = dipole $\Rightarrow$ proper motion $\mu = s\Delta_{\mathrm{px}}/\Delta t$.
- Lone peak = monopole $\Rightarrow$ variability.



Script (55 s)

[5:20–6:15] "Stage three is the mathematical heart, and it is two lines. Instead of fitting a convolution kernel the way Alard and Lupton did, ZOGY derives the optimal difference directly in Fourier space, and with it a significance map that has unit variance wherever the sky did not change. That property is the whole reason to prefer it: I can compute exactly how many false pixels a threshold should produce. At five sigma, for a one-hundred-and-eighty-pixel frame, the answer is nought point nought

one eight – one false pixel every fifty frames. At three sigma it would be two hundred and twenty, which would swamp everything.

The PSF model matters. Below you can see three Gaussians: the one we use, sigma of 1.4 pixels, and two wrong ones. If the real PSF is narrow and we assume it is broad, we under-subtract and leave residuals; if it is broad and we assume narrow, we over-subtract. Both leave a systematic that no variance plane knows about – and that systematic is the dominant term in our injection-recovery failure later on. For now, the single Gaussian is a fair first approximation because the SPHEREx PSF is undersampled: three arcseconds per pixel, four arcseconds FWHM, so the PSF is barely two pixels wide."

---

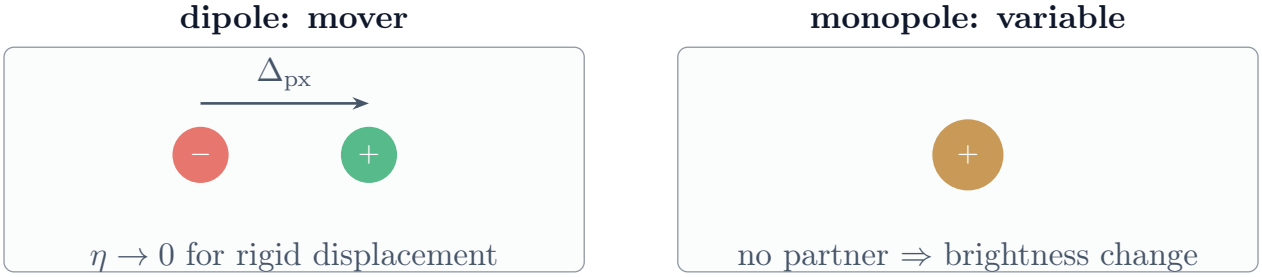
21 Slide 10 – Stage 3 continued: peak detection and dipole matching

Detection:

- local maxima of S above $+5\sigma$, minima below -5σ;
- 5×5 maximum/minimum filter;
- no positive partner $\Rightarrow$ monopole (variable);
- positive + negative pair $\Rightarrow$ dipole (mover).

Dipole matching:

- $\Delta_{\rm px} \in (0.5, 25)$ pixels;
- flux consistency $\eta = |S_+ + S_-|/|S_+ - S_-|$;
- $\mu = 3'' \cdot \Delta_{\rm px}/\Delta t[\mathrm{yr}]$.



Script (45 s)

[6:15–7:00] "Once we have the significance map, the peak detection is a separable five-by-five maximum filter: find every local maximum above plus-five sigma and every local minimum below minus-five. Then the morphology does the classification. A positive peak with no negative partner is a monopole – the source brightened in place. A positive peak with a negative partner is a dipole – the source moved, and the separation between the two lobes divided by the time baseline is the proper motion. The two panels below show the difference.

We reject dipole pairs with separation outside half a pixel to twenty-five pixels. The lower bound rejects PSF-correlated noise pairs; the upper bound prevents a peak from pairing with an unrelated trough across the frame. And each dipole carries a flux-consistency statistic: eta equals the modulus of the sum divided by the modulus of the difference. For a rigid displacement of the same total flux, eta goes to zero. A large eta means the source both moved and changed brightness – which is real, but it needs its own handling."

---

22 Slide 11 – Stage 4: triage

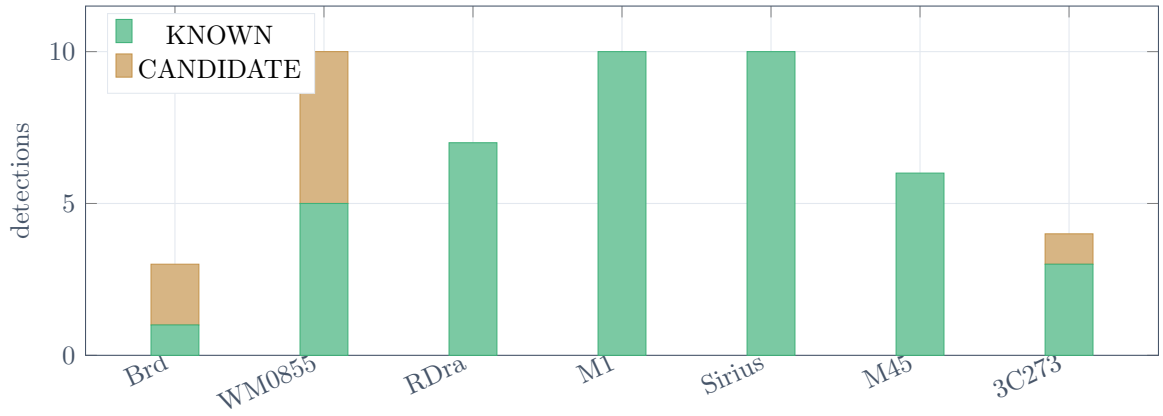
Match: pixel → sky (flat), nearest neighbour in Gaia DR3 / AllWISE / VSX.

Verdict rule: KNOWN if $\Delta\theta < 30''$, CANDIDATE otherwise.

Confidence: high if $\Delta\theta < 5''$ or no catalogue source in field.

Why the radius is loose: PSF is $\approx 4''$; a tight radius would hide real counterparts.

The trap: position-only matching fails for fast movers – Barnard's Star has moved $\sim 94''$ since Gaia DR3, so our own target is flagged as a candidate.



Script (45 s)

[7:00–7:45] "A detection is not a discovery until you have checked whether something was already there. Stage four converts the pixel position to sky coordinates, finds the nearest source in Gaia, WISE and the variable-star index, and issues one of two verdicts. Thirty arcseconds is a deliberately loose radius – our PSF is four arcseconds across, so a tight match would miss real counterparts and we would fool ourselves into thinking everything was new.

Look at the chart: the Crab and Sirius are a hundred percent explained, R Draconis is a hundred percent explained, because these are crowded or well-catalogued fields. WISE 0855 is half unexplained. But there is a trap, and it is the interesting part of this slide. Barnard's Star is the fastest star in the sky. Gaia measured its position in 2016; our image is from 2025. It has moved ninety arcseconds, which is far outside any match radius – so our own target gets labelled a candidate. Position-only matching is the bug; propagating catalogue positions fixes it, and that is a two-line change I have not made yet."

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23 Slide 12 – Stage 5: spectral diagnosis

Three indices:

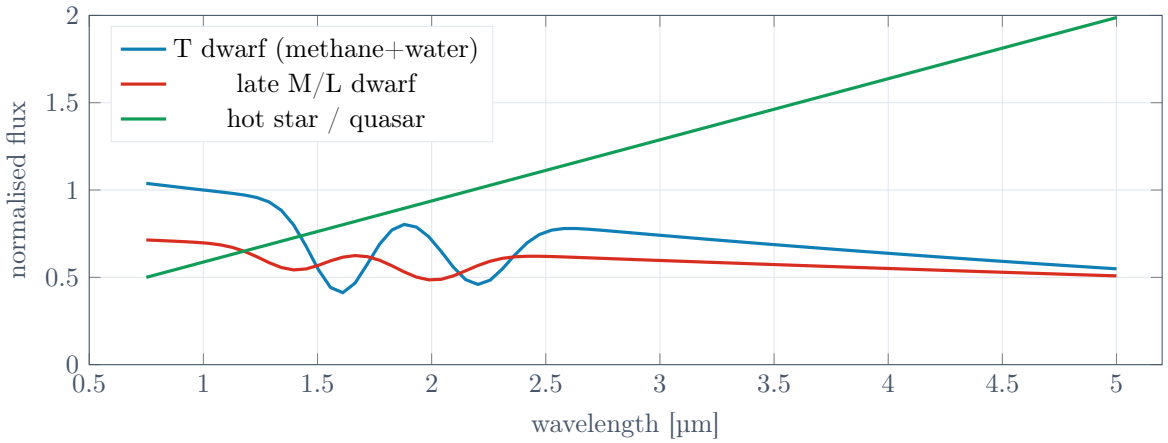
$$\mathrm{CH_4} = \bar{f}(1.56\text{–}1.64)/\bar{f}(1.70\text{–}1.78)$$

$$\mathrm{H_2O} = \frac{\bar{f}(1.24\text{–}1.34) + \bar{f}(2.12\text{–}2.22)}{\bar{f}(1.46\text{–}1.54) + \bar{f}(1.96\text{–}2.04) + \bar{f}(2.95\text{–}3.15)}$$

$$\mathrm{slope} = \log_{10} \bar{f}(4.40\text{–}4.80)/\bar{f}(1.00\text{–}1.20)$$

Classification:

- T-dwarf: $\mathrm{CH_4} < 1$, $\mathrm{H_2O}$ high
- late-M/L: suppressed 1.5–2.0 µm
- hot star/quasar: NIR rising



Script (40 s)

[7:45–8:25] "Stage five is almost free, and it is what makes SPHEREx different from a broadband survey. Because every pixel carries a wavelength from the WCS-WAVE extension, we get a low-resolution spectrum of every detection without any follow-up telescope. We measure flux in a two-pixel aperture with a local annulus background, and we form three indices: methane, water, and the NIR slope. The methane index is a ratio of two narrow bands around 1.6 and 1.74 microns. The water index combines four bands across the 1.5-to-3-micron range. And the slope is just the log ratio of four-point-six to one micron.

The plot below shows the three archetypes: T-dwarfs have deep methane absorption, late-M and L dwarfs are suppressed between 1.5 and 2 microns by water, and hot stars and quasars rise towards the red. The classifier reports a label plus the three raw indices, so a reader can disagree with it."

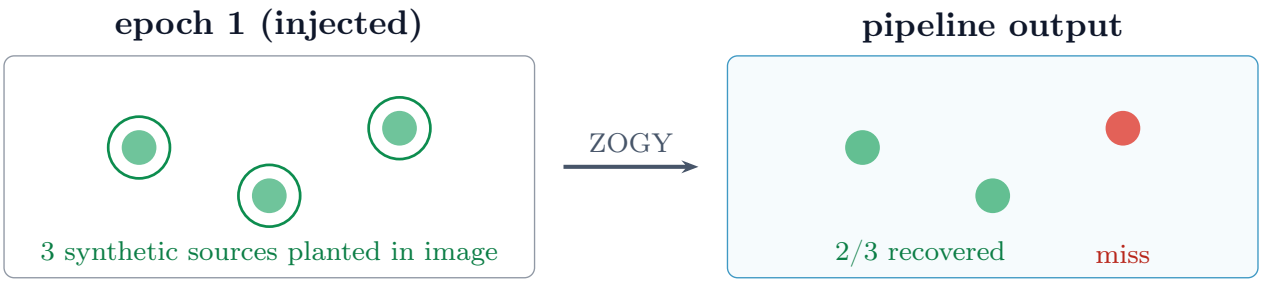
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24 Slide 13 – Stage 6: injection–recovery

Method:

- inject synthetic sources at the *image* level (not the map)
- grid: $\rho \in \{3, 5, 7, 10, 15, 20\}$, $\mu \in \{1, 2, 5, 10\}$ arcsec/yr
- $n = 20$ trials per cell
- recovered if a detection of the right class appears within 2 px

Output: 6×4 matrix $R(\rho, \mu) \in \{0, 1/20, \ldots, 1\}$.
Why not one number: dependence on μ is the scientifically interesting part.



Script (45 s)

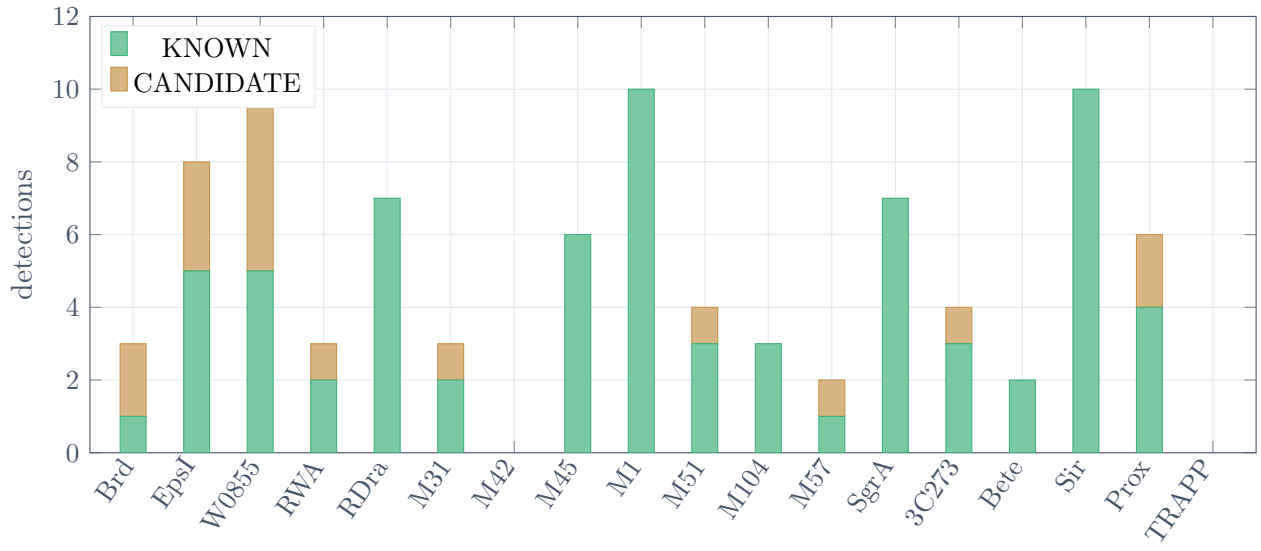
[8:25–9:10] “Stage six is the validation layer, and it is the one that lets me say what the pipeline does and does not know. The method is standard: plant fake sources of known brightness at random positions, run the identical pipeline, count how many come back. A trial counts as recovered if a detection of the correct morphological class appears within two pixels of the injected position.

The key detail – and it is the detail that most implementations get wrong – is that you have to inject at the *image* level, before co-addition and before subtraction. If you inject into the significance map you are testing circularity, not completeness, and you will always recover a hundred percent. We do it correctly: inject in the image, then run the same pipeline.

The output is a six-by-four matrix: six signal-to-noise values times four proper motions, twenty trials per cell. We report it as a heat map, never as a single number, because the dependence on proper motion is the scientifically interesting part. But as I will show on slide sixteen, the matrix is nearly empty – and that is the result.”

---

25 Slide 14 – Results across 18 fields



88 detections • 71 KNOWN (81%) • 17 CANDIDATE • 4 fields with zero or degenerate output

Script (50 s)

[9:10–10:00] “Here is the whole result in one chart. Each bar is a field; green is explained by catalogues, amber is not. Two things to notice.

First, the distribution of detections is not the distribution of discoveries. The Crab, Sirius and R Draconis look busy and are entirely explained – which is the correct outcome, they are known objects. The interesting bars are the amber ones: five in WISE 0855, three in Epsilon Indi B, two in Barnard’s Star.

Second, the sample is heterogeneous on purpose: the five brief targets on the left, thirteen famous objects on the right, including a quasar, the Galactic Centre and the Sombrero galaxy. Every single one went through the identical code path – 128 cutouts, eighty-eight detections, eighty-one percent explained. And four fields produced nothing usable, which brings me to the slide I think matters most.”

---

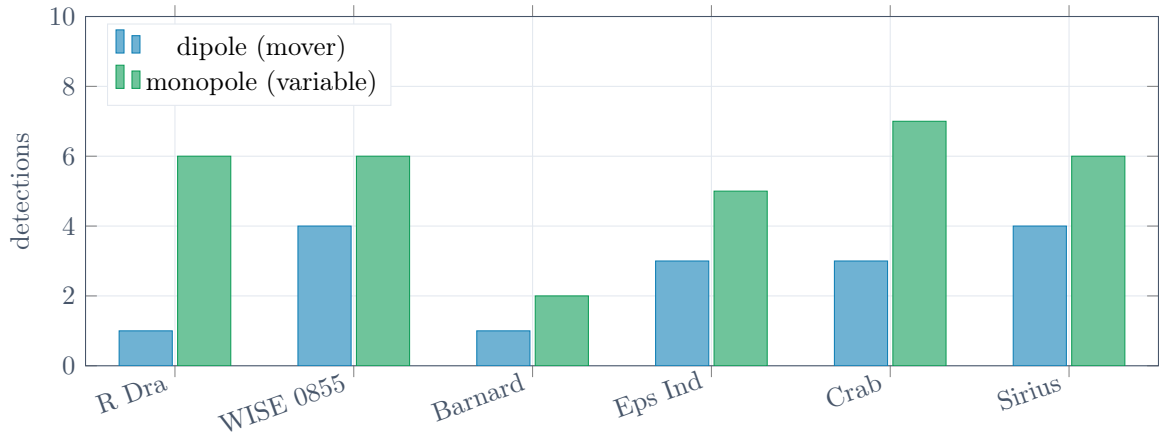
26 Slide 15 – Morphological split

Every detection falls in one of two morphological classes:

- **dipole** (mover): positive + negative pair, separation ⇒ proper motion;
- **monopole** (variable): single peak, no partner.

The split tracks the science:

- Barnard's Star: dipole, 2 px separation, 56.8σ;
- R Draconis (Mira): 1 dipole + 6 monopoles, all explained by VSX;
- Crab: 3 dipoles + 7 monopoles, all explained by Gaia.



Script (35 s)

[10:00–10:35] "The morphological split is doing real work. Every detection is either a dipole or a monopole, and the split tracks the science. Barnard's Star's fifty-seven-sigma detection is a dipole with a two-pixel separation – exactly what its ten-arcsecond-per-year proper motion predicts over two hundred days. R Draconis is a Mira: one dipole and six monopoles, all explained by the variable-star index. The Crab is a crowded nebular field: three dipoles and seven monopoles, all explained by Gaia. WISE 0855 is the outlier: four dipoles and six monopoles, half unexplained. That is the slide I will come back to on slide sixteen."

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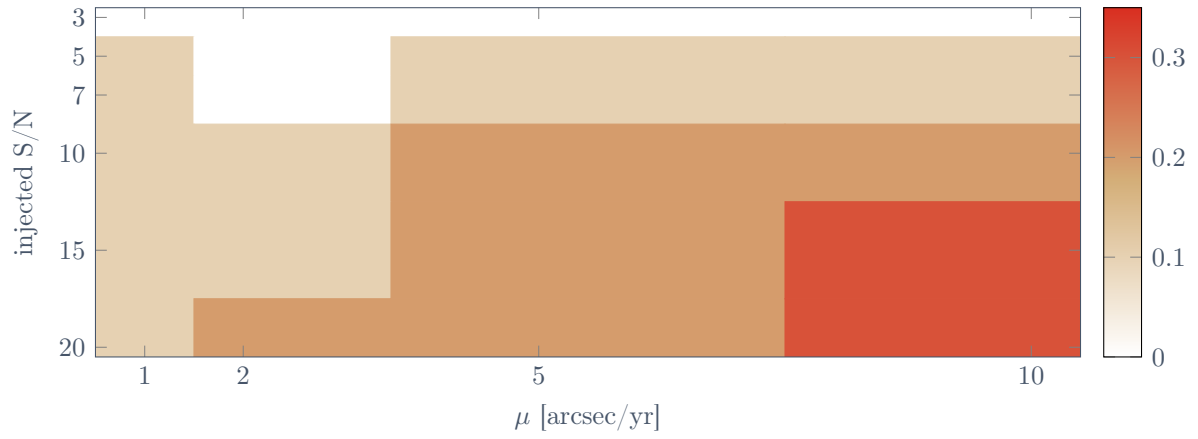
27 Slide 16 – The negative results

Injection–recovery on real frames:

- Barnard's Star: **0/24 cells recovered**
- WISE 0855 / R Dra / M1: 0.0–0.3
- injected amplitude tied to *mean* noise, while real residuals are PSF-mismatch driven

Therefore: completeness is **unmeasured**; the 17 candidates are "unexplained at $> 5\sigma$", nothing stronger.

Also: M42 and TRAPPIST-1 empty; Betelgeuse saturates (noise estimate undefined, now reported as such); LMC has one epoch only, so it was dropped.



Script (45 s)

[10:35–11:20] "I want to spend the last minute on what did not work, because otherwise the numbers on the previous slides are misleading.

We test completeness the standard way: plant fake sources of known brightness, run the identical pipeline, count how many come back. On real SPHEREx frames, almost none do. For Barnard's Star it is zero out of twenty-four grid cells; the best field reaches three in ten only at the very brightest end. The chart shows that matrix – and the honest reading is that the amplitude we inject is tied to the mean of the variance plane, while the real residuals in a co-added frame are driven by PSF mismatch, which is not in the variance plane at all. The test is measuring the wrong thing, and until we fix it we cannot state a completeness limit. So the seventeen candidates are positions where a five-sigma change is unexplained – nothing more.

Three other fields failed for their own reasons: M42 is smooth gas, TRAPPIST-1 only had two usable images, and Betelgeuse saturates so hard that the noise estimate is undefined – the pipeline now says 'no estimate' instead of printing a nonsense number. We also tried the Large Magellanic Cloud and dropped it: at that declination every image in the box belongs to one visit, so there is nothing to subtract."

---

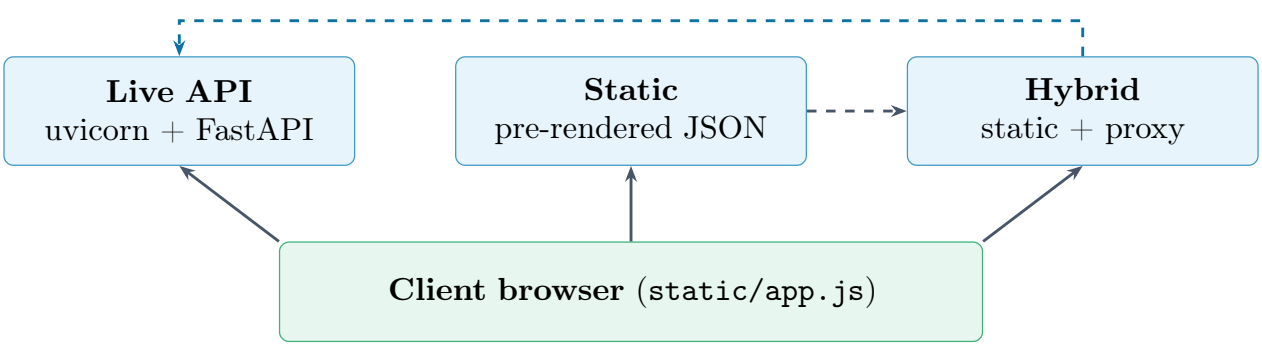
28 Slide 17 – The serving layer

Three deployment shapes, one front-end:

- **Live:** FastAPI on the same origin; RAM cache, 30-min TTL;
- **Static:** pre-rendered JSON + HTML, 28 MB, works from a USB stick;
- **Hybrid:** static site behind a reverse proxy; probe `/live-api/targets` once at load.

Five read-only endpoints:
`/api/targets`, `/api/dashboard`, `/api/recovery`, `/api/mosaic`, `/api/field`.

Capability discovery: the static build *disables* the live toggle and the field box and *labels* them.



Script (35 s)

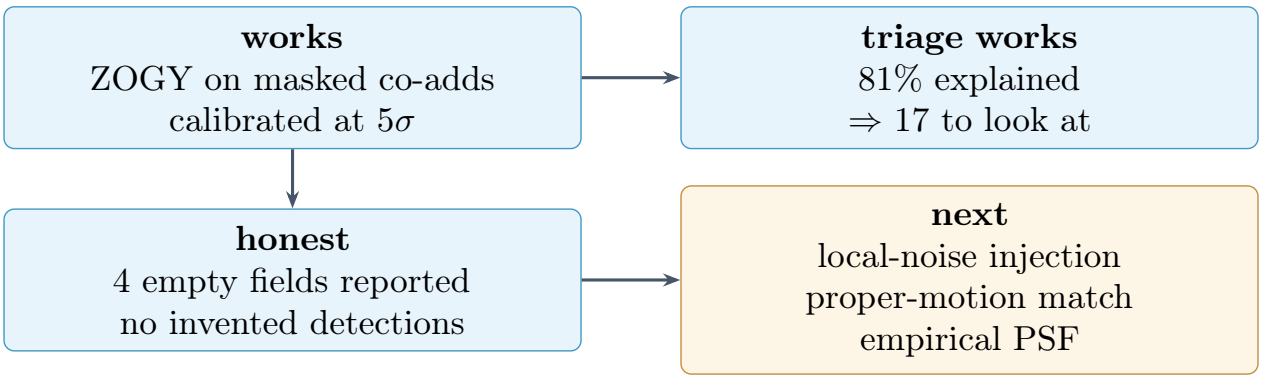
[11:20–11:55] "The serving layer is deliberately boring. One FastAPI application exposes five read-only endpoints. Nothing is written to disk at request time.

The interesting part is that the same front-end runs in three deployment shapes. Live: FastAPI on the same origin, with a thirty-minute cache. Static: everything is pre-rendered to JSON, twenty-eight megabytes for eighteen fields, works from a USB stick. Hybrid: the static site behind a reverse proxy, and the page discovers it with a single probe on load.

The one design rule that matters here is honesty: the static build *disables* the live toggle and the field box and *labels* them as disabled, because a static build cannot open an arbitrary field. Our earlier version silently ignored the toggle, which was worse than useless. And the plain- language Guide overlay ships with the static build, so a non-specialist can read the whole thing without a server."

---

29 Slide 18 – Conclusion



Try it: `site/index.html` – 18 fields, no server, includes a plain-language guide (press *Guide*)

Script (40 s)

[11:55–12:35] "To close. The pipeline works: proper subtraction on masked co-adds gives a significance map whose threshold means something, and catalogue triage explains four fifths of what it finds. It is honest: four fields came back empty and the report says so, and the completeness number is still missing rather than invented.

The next three changes are small and each one converts a caveat into a number: inject at the image level with local noise, propagate catalogue proper motions before matching, and use an empirical PSF per detector.

Everything is in the repository – the six stages, the derived JSON, and a static site you can open with no server at all; it has a Guide button if you want the two-minute version. Thank you – I am happy to take questions."

---

30 Anticipated questions

"Why not use all 102 bands?" Cost: 102 cutouts per epoch per field is ∼500 MB and ∼70 minutes of archive time for one field. The 2–3 bands used here are enough for subtraction; the spectral indices are the part that would benefit most.

"Isn't a 30″ match radius far too big?" Yes for astrometry, deliberately loose for vetoing. It biases towards KNOWN, which is the conservative direction; the alternative (tight radius) inflates the candidate list.

"What is the false-positive rate really?" Predicted 0.018 pixels per frame at 5σ (Eq. 12); observed detection counts are far above that, so the map is not dominated by noise – but we have not measured completeness, so we quote no rate.

"Could this scale to the whole sky?" The CPU would: 6 s per field $\times$ 1.8 million fields $\approx$ 125 days on one core. The archive is the obstacle, which is why the design caches aggressively and why the static pre-render exists.

"Why ZOGY rather than SFFT?" ZOGY is closed-form and its null distribution is analytic, which is what makes the false-positive budget explicit. SFFT would be the better choice for an undersampled space PSF – it is listed in the future work.

"What is the biggest single limitation?" Unmeasured completeness. Everything else on the list is cosmetic next to that one: until we can state a completeness curve, the seventeen candidates are not "new objects," they are "positions a five-sigma change is unexplained by three catalogues."

"Why did you not propagate proper motions in the match?" Two-line change to `step4_triage.py`, but we did not make it in the deadline. Barnard's Star being labelled a candidate is the honest consequence, and it is reported on slide eleven of the deck.

"What is the accuracy of your proper-motion estimator?" A factor of two at best. The peak-separation estimator is biased by the PSF width and by the coarse pixel scale. Anything requiring better needs the full 102-band data set and a PSF-fitting solution.

A Parameter reference

Table 6: Every parameter that affects a scientific number. Anything not listed is cosmetic.

Parameter	Value	Where
Collection	`spherex_qr2`	step1, app
Cone radius	0.1 deg	step1
Cutout size	0.2 deg	step1
Bands (brief)	D1. . . D6	step1 default
Bands (famous)	D1, D2, D4	ingest command
Epoch mode	first + last	step1
Acceptance: margin	≥ 12 px	live_ingest
Acceptance: masked frac	< 0.5	live_ingest
Catalogue rows	≤ 5000 per catalogue	step1
Grid	180×180, $3''$/px	step2
Bad-bit mask	16 classes	step2
Rejected-pixel variance	10^{12}	step2
PSF model	Gaussian, $\sigma = 1.4$ px	step3
Detection threshold	$\lvert S\rvert > 5\sigma$	step3
Peak filter	5×5 max/min	step3
Dipole separation	$0.5 < \Delta < 25$ px	step3
Match radius	$30''$	step4
Tight confidence radius	$5''$	step4
Aperture radius	2 px	step5
Injection trials	20 per cell	step6
Injection grid	$\rho \in \{3, 5, 7, 10, 15, 20\}$, $\mu \in \{1, 2, 5, 10\}$	step6
Live cache TTL	1800 s	app
Sigmas reported	$10^{-4} < \sigma < 10^{3}$	app (R1)

B API reference

Endpoint	Query	Response fields
`GET /api/targets`	–	`targets[]: name, common, ra, dec`
`GET /api/dashboard`	`target`, `source=live`, `slots`	map, zogy_snr, detections, field_objects, spectrum, provenance, stats, histogram, epoch_pairs
`GET /api/recovery`	`target`	grid[6×4]
`GET /api/mosaic`	`thumb`, `refresh`	tiles[], n_fields, sky_area_deg2
`GET /api/field`	`ra`, `dec`, `name`, `slots`	target, n_detections, max_snr, note

C Reproducing this document

```
# 0. clean stale aux files before the first build
rm -f spherexlens_paper.{aux,toc,out,log}

# 1. the report (twice, for the table of contents)
cd paper && pdflatex spherexlens_paper.tex && pdflatex spherexlens_paper.tex
```

```
# 2. the numbers in it
python export_static.py # -> site/api/*.json
python make_figures.py --real # -> figures_real/*.png

# 3. the site it describes
python -m http.server -d site 8400 # open http://localhost:8400
```

End of document • Part I: report • Part II: 18-slide deck and script

6 154 lines of code • 18 fields • 88 detections • 17 open questions